

PRIMARY 1 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Breakdown (Subtopics)
1	Whole Numbers 1–5	- Counting objects from 1 to 5 - Recognition and identification of numbers 1–5 - Writing numbers 1–5 correctly - Matching numbers to quantities (1–5) - Ordering numbers 1–5 from smallest to largest
2	Whole Number 0 (Zero)	- Introduction to the concept of zero (nothing/empty) - Recognition and identification of the number 0 - Writing the number 0 correctly - Understanding zero as an empty set - Counting from 0 to 5
3	Whole Numbers 6–9	- Counting objects from 6 to 9 - Recognition and identification of numbers 6–9 - Writing numbers 6–9 correctly - Matching numbers to quantities (6–9) - Ordering numbers 1–9 from smallest to largest
4	Whole Number 10	- Introduction to number 10 - Counting objects from 1 to 10 - Recognition, identification and writing of number 10 - Understanding 10 as a complete set - Ordering numbers 0–10
5	MID-TERM EXAMINATION	Assessment of Weeks 1–4
6	MID-TERM BREAK	Holiday/Rest
7	Addition (Introduction)	Understanding addition as "putting together" or

		"adding more" - Using objects to demonstrate addition (1+1, 2+1, etc.) - Introduction to the plus sign (+) and equals sign (=) - Simple addition within 5 (vertical and horizontal format) - Word problems on addition within 5
8	Addition (Continued)	- Addition within 10 using objects and pictures - Adding zero to any number - Addition facts for numbers 1–10 - Word problems on addition within 10 - Practical activities: adding toys, fruits, pencils, etc.
9	Subtraction (Introduction)	- Understanding subtraction as "taking away" or "removing" - Using objects to demonstrate subtraction - Introduction to the minus sign (–) - Simple subtraction within 5 (vertical and horizontal format) - Word problems on subtraction within 5
10	Subtraction (Continued)	- Subtraction within 10 using objects and pictures - Subtracting zero from any number - Subtraction facts for numbers 1–10 - Word problems on subtraction within 10 - Practical activities with real objects
11	REVISION	Review of all topics: Numbers 0–10, Addition and Subtraction
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends

PRIMARY 1 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS 1--5

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count objects from 1 to 5 clearly.
2. Identify and read the numbers 1, 2, 3, 4, and 5.
3. Write the numbers 1 to 5 correctly.
4. Match numbers to the correct quantity of objects (e.g., matching the number 3 to three balls).
5. Order the numbers 1 to 5 from smallest to largest.

CONTENT

Counting Numbers



1. Counting Objects from 1 to 5

- Counting tells us "how many." It is a way of finding out the total number of things in a group.
- We must touch or point at each object as we say its number name. This is called **one-to-one correspondence**.
- **Practice:**
 - Let's count our **fingers**: One (touch thumb), Two (touch index), Three, Four, Five.
 - Let's count **objects in the classroom**: 1 door, 2 windows, 3 chairs, 4 tables, 5 books.
- **Counting Rhyme**: "One, two, three, four, five, once I caught a fish alive!"

2. Recognition and Identification of Numbers 1--5

- Numbers have names and shapes. We must learn to recognise them like we recognise letters.
- Here are some fun ways to remember the shapes:
 - **1** looks like a straight stick or a pencil standing up.
 - **2** looks like a sitting duck with a curved neck.
 - **3** looks like an open ear or two bumps on a slide.
 - **4** looks like a flag on a tall pole or a chair turned over.
 - **5** has a flat hat on its head and a big tummy.

3. Writing Numbers 1--5

- Writing numbers neatly is important. We start at the top and go down.
- **Correct Formation:**
 - **1**: One straight stroke down from top to bottom.
 - **2**: Start at the top, curve round to the left, then dash straight to the right.
 - **3**: Start at the top, curve right, then curve right again (like two bumps).
 - **4**: Stroke down, dash right, then a straight stroke down through the middle.
 - **5**: Draw a small dash on top, curve down and round, then a flat dash at the bottom.

- We practice writing in the air first, then on a friend's back, then on paper.

4. Matching Numbers to Quantities

- This skill links the **number symbol** (3) with the **actual amount** (ooo).
- **Examples from Daily Life:**
 - One nose on your face. → 1
 - Two eyes on your face. → 2
 - Three wheels on a tricycle. → 3
 - Four legs on a chair. → 4
 - Five fingers on one hand. → 5
- **Matching Activity:** Pupils draw lines to connect a group of objects (e.g., four stars) to the correct numeral (4).

5. Ordering Numbers from Smallest to Largest

- **Smallest** means the least amount. **Largest** means the most amount.
- We can think of numbers like steps on a ladder. Number 1 is the first, low step. Number 5 is the highest step.

Number Line (Ladder):

5 (Top step) -----
 4 -----
 3 -----
 2 -----
 1 (First step) -----

- The correct order from smallest (least) to largest (most) is: **1, 2, 3, 4, 5**.

EVALUATION

1. Count how many fingers are on your right hand.
2. Write the number **2**.
3. Draw **3** balls in your book.

4. Which number comes immediately after 4?
5. Arrange these numbers from the smallest to the biggest: 3, 1, 5, 2, 4.

ASSIGNMENT

1. Write the numbers **1, 2, 3, 4, 5** three times each in your exercise book.
2. Draw **4** cups and write the number "**4**" next to them.
3. Count the number of chairs in your dining room at home (count up to 5 only).



WEEK 2: WHOLE NUMBER 0 (ZERO)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define zero as "nothing" or "empty."
2. Identify the number symbol 0.
3. Write the number 0 correctly.
4. Count from 0 to 5 in the correct order.

CONTENT



Zero Means Empty



Empty wallet



Empty box



Empty plate



Empty cage



No cars



Dead battery

1. Introduction to Zero

- **Zero (0)** is a special number. It means there is **nothing** or **none**.
- It represents an **empty set**—a group with no objects in it.
- **Real-life Examples:**

- If you have 2 biscuits and you eat both, you have **0** biscuits left.
- An empty plate has **0** pieces of food on it.
- A parking lot with no cars has **0** cars.

2. Recognition of Number 0

- The number **0** looks like a circle or an egg. It is round with no corners.
- It is the only number that is shaped like a closed loop.
- We can find **0** on telephones, remote controls, and clocks.

3. Writing the Number 0

- To write **0**, start at the top, curve your pencil all the way around to the left, go down, and close the circle back at the top.
- It is like drawing a perfect circle. Practice by tracing the rim of a bottle cap.

Start → ○ → Finish
(Top)

4. Understanding Zero as an Empty Set

- **Activity:** Show a clear jar with 3 marbles. Take out one marble at a time, counting down: "3... 2... 1... 0." When the jar is empty, we say it contains **0** marbles.
- This helps us understand that **0** is a real number that tells us about "nothingness."

5. Counting from 0 to 5

- Now that we know **0**, it becomes the **starting point** of our counting.
- The correct order when we begin from nothing is: **0, 1, 2, 3, 4, 5**.
- **Visual Aid:** A number line starting at 0.

0---1---2---3---4---5

Start here

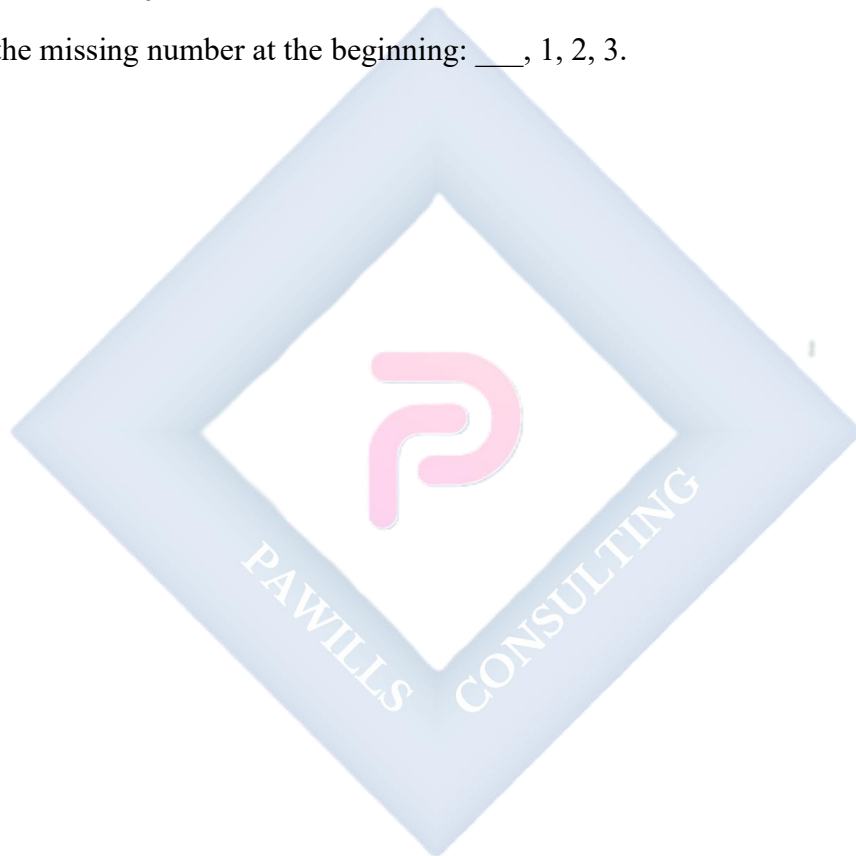
EVALUATION

1. What number means "nothing" or "empty"?

2. Write the number **0**.
3. If a tree has no apples on it, how many apples are there?
4. What number comes **before** 1?
5. Count aloud from 0 to 5.

ASSIGNMENT

1. Draw a picture of an **empty** basket and write the number **0** next to it.
2. Write the number **0** five times in a row.
3. Fill in the missing number at the beginning: __, 1, 2, 3.



WEEK 3: WHOLE NUMBERS 6--9

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count objects from 6 to 9.
2. Identify and write the numbers 6, 7, 8, and 9.
3. Match the numbers 6--9 to the correct quantity of objects.
4. Order numbers 1 to 9 from smallest to largest.

CONTENT



1. Counting Objects from 6 to 9

- We are building on our counting skills. For numbers 6-9, we must count carefully, as the groups get bigger.
- **Strategy:** Count out loud and move each object as you count it to avoid counting the same thing twice.
- **Practice with Counters:**

- Count out **6** bottle caps.
- Make a line of **7** pencils.
- Gather **8** leaves.
- Find **9** pebbles.

2. Recognition and Writing of Numbers 6--9

- Each number has a unique shape. We can remember them with fun pictures:
 - **6:** Curve down and curl it in at the bottom, like a whistle or a snake curling up.
 - **7:** A straight dash to the right, then a slanted stroke down to the left, like a walking stick.
 - **8:** Make an 'S' shape and then close it at the top, like a snowman or a pair of glasses.
 - **9:** First draw a circle at the top, then add a straight line down, like a balloon on a string.
- **Air Writing:** Practice forming these numbers in the air with big arm movements.

3. Matching Numbers to Quantities (6-9)

- This requires careful counting. For example, if you see the number **7**, you should be able to count out 7 objects.
- **Everyday Connections:**
 - **6:** The number of sides on a die (cube).
 - **7:** The number of days in a week.
 - **8:** The number of legs on a spider.
 - **9:** The number of players in a baseball team.

4. Ordering Numbers 1--9

- Now we have a longer number ladder. We must arrange all numbers from the baby (1) to the big brother (9).
- The correct **ascending order** (smallest to largest) is: **1, 2, 3, 4, 5, 6, 7, 8, 9.**

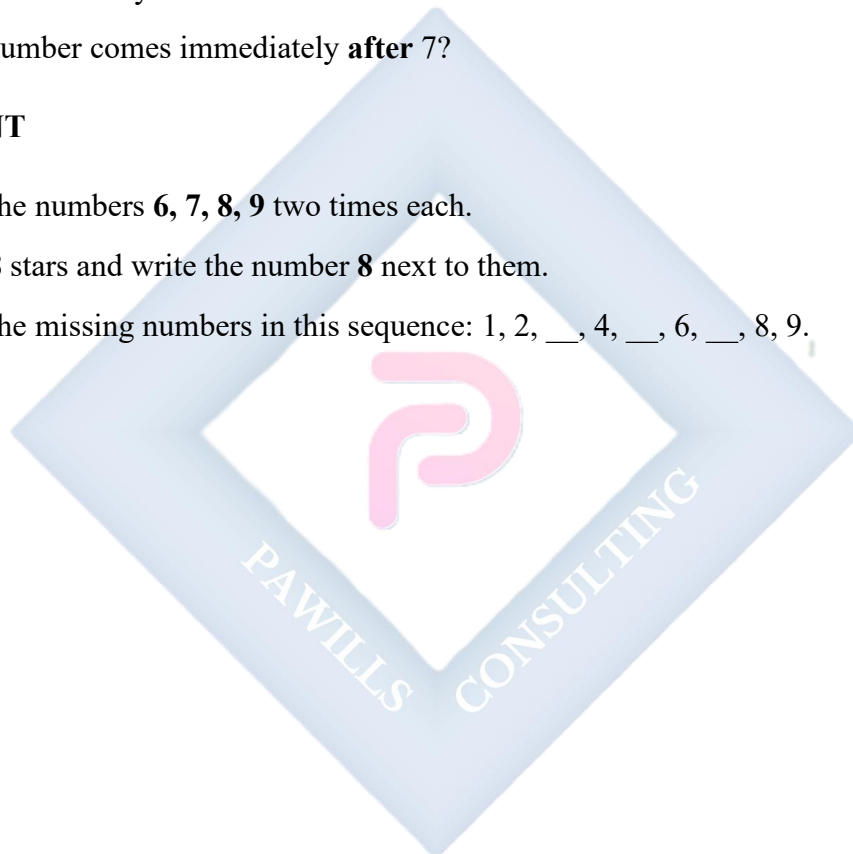
- **Game:** Give pupils cards with numbers 1-9. Let them arrange themselves in a line in the correct order.

EVALUATION

1. Count out **7** of your crayons and show the teacher.
2. Write the number **8**.
3. Which number represents a larger amount: 6 or 9?
4. Draw **6** circles in your book.
5. What number comes immediately **after** 7?

ASSIGNMENT

1. Write the numbers **6, 7, 8, 9** two times each.
2. Draw **8** stars and write the number **8** next to them.
3. Write the missing numbers in this sequence: 1, 2, __, 4, __, 6, __, 8, 9.



WEEK 4: WHOLE NUMBER 10

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the number 10 as a two-digit number.
2. Count objects up to 10.
3. Write the number 10 correctly.
4. Order numbers from 0 to 10.

CONTENT



1. Introduction to Number 10

- The number **10** is a milestone! It is our first **two-digit number**. It is made of two symbols: a **1** and a **0**.
- It represents a **complete set** or a **full bundle**. For example, we have 10 fingers in total (5 on each hand).

- In many counting systems, reaching 10 is a reason to start a new group.

2. Counting Objects 1 to 10

- We can now count all the way to ten. Let's count together slowly and clearly.
- **Activity:** "Let's build a tower of 10 blocks." Count each block as you add it: 1, 2, 3... 10.
- **Tip for 10:** When you have 10 items, you can often group them into two fives (5 and 5) to check.

3. Writing Number 10

- Writing **10** is a two-step process:
 - a. First, write the digit **1** (a straight stick).
 - b. Immediately to its right, write the digit **0** (a circle).
- Make sure the **1** and the **0** are close together but not touching, so it looks like "10" and not "1 0".

4. Understanding 10 as a Complete Set

- **10** feels "complete." Think of a full set of 10 crayons, 10 toes, or 10 eggs in a carton.
- In our number system, after 9, we start a new sequence with 10, 11, 12, etc. Ten is the foundation.

5. Ordering Numbers 0--10

- Let's put all the numbers we have learned in order, from the very beginning (0) to our new big number (10).
- The correct sequence is: **0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.**
- **Visual Number Line:**

0-1-2-3-4-5-6-7-8-9-10

EVALUATION

1. How many fingers do you have on both hands put together?
2. Write the number **10**.

3. Which two digits (numbers 0-9) are used to write the number ten?
4. Count aloud from 1 all the way to 10.
5. Which is the biggest number among these: 8, 5, or 10?

ASSIGNMENT

1. Write the numbers from **0 to 10** in the correct order in your book.
2. Draw **10** lollipops (or circles) in a row.
3. Look at this list: 1, 0, 10, 7, 2. Circle the number **ten**.



WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Week 1 to Week 4.

CONTENT

Assessment of Weeks 1–4

- Counting and writing numbers **0 through 10**.
- Matching a given number of objects (up to 10) to its numeral.
- Ordering a set of numbers from 0 to 10 from smallest to largest.

EVALUATION

- Administer the written and oral Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

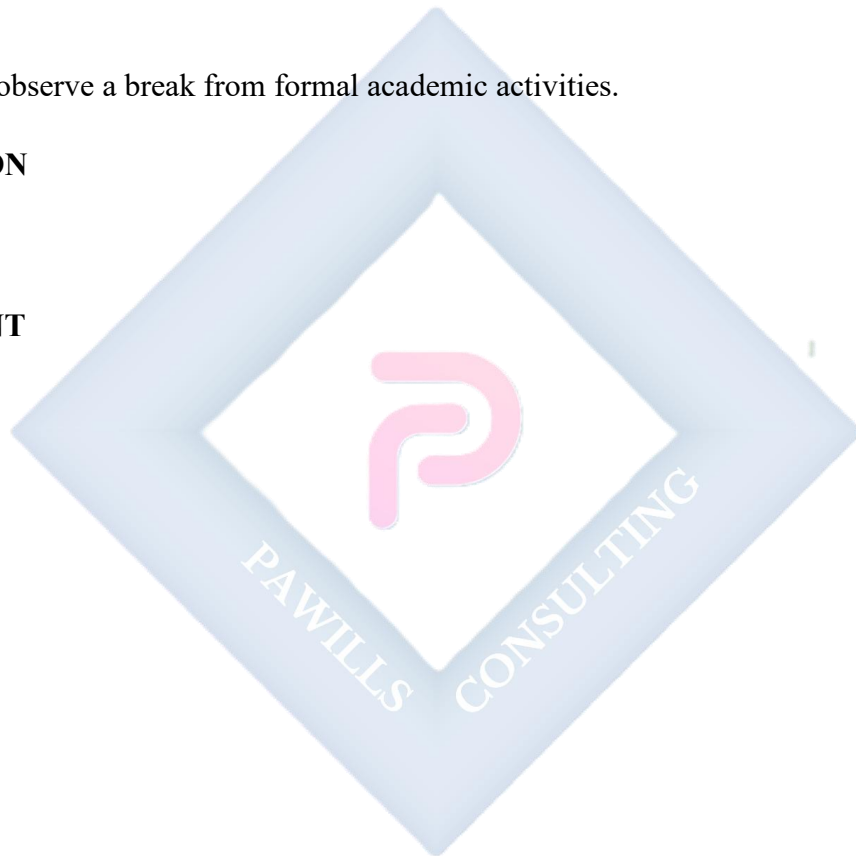
- Pupils observe a break from formal academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



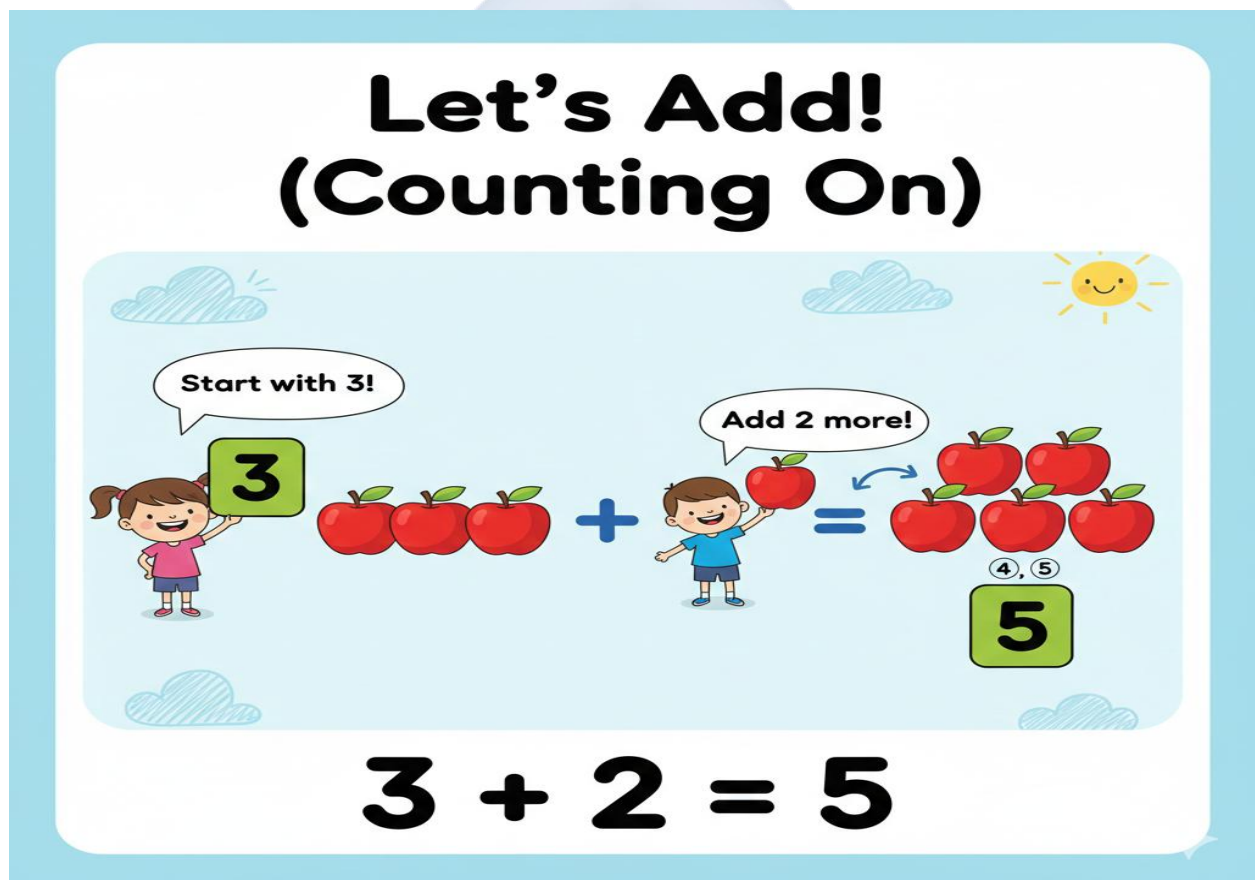
WEEK 7: ADDITION (INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define addition as "putting things together."
2. Identify the plus sign (+) and equals sign (=).
3. Add two single-digit numbers where the sum is not more than 5.
4. Solve simple word problems involving addition up to 5.

CONTENT



1. Understanding Addition

- **Addition** is the process of combining two or more groups to find the **total** or **sum**. It answers the question, "How many altogether?"

- We use words like **add**, **plus**, **combine**, and **altogether**.
- **Concrete Example:** "You have 2 pencils in your hand. I give you 1 more pencil. If you put them together, how many pencils do you have now?"

2. Signs for Addition

- **The Plus Sign (+):** This symbol tells us to add or combine amounts. It looks like a cross.
- **The Equals Sign (=):** This symbol means "is the same as" or "the answer is." It shows that what is on the left (e.g., $2+1$) has the same value as what is on the right (e.g., 3).

Example: $2 + 1 = 3$

Group 1	Add	Group 2	Total

3. Simple Addition Within 5

- We start with small numbers. Use your fingers or physical objects (counters) to help.
- **Process:** Start with the first number, then **count on** the second number.
 - **2 + 1:** Think "2... then count on **one** more: 3." So, $2 + 1 = 3$.
 - **3 + 2:** Think "3... count on two: 4, 5." So, $3 + 2 = 5$.
- **Vertical Addition Practice:**

$$\begin{array}{r} 2 \\ +1 \\ \hline 3 \end{array}$$

4. Word Problems Involving Addition

- A word problem tells a short story with numbers. We must find the numbers and add them.
- **Example:** "Kemi has 2 sweets. Her brother gives her 2 more sweets. How many sweets does Kemi have now?"
 - Find the numbers: 2 and 2.

- Key word: "more" suggests adding.
- Solve: $2 + 2 = 4$. Kemi has 4 sweets.

EVALUATION

1. In your own words, what does **addition** mean?
2. Draw the **plus sign** and the **equals sign**.
3. What is $1 + 1$?
4. Calculate: $2 + 2$.
5. Solve this problem: "I have 3 oranges and I find 1 more on the table. How many oranges do I have altogether?"

ASSIGNMENT

1. Solve this: $2 + 1 = ?$
2. Solve this: $3 + 2 = ?$
3. Draw 2 apples and then draw 2 more apples next to them. Write the addition sentence and the total (e.g., $2 + 2 = 4$).

WEEK 8: ADDITION (CONTINUED)

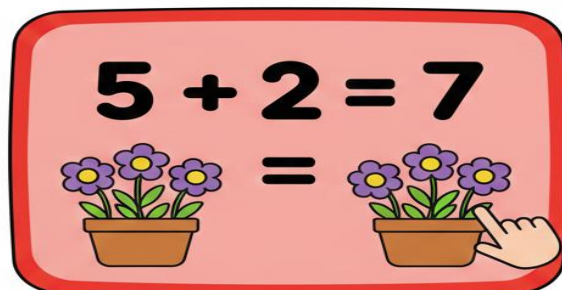
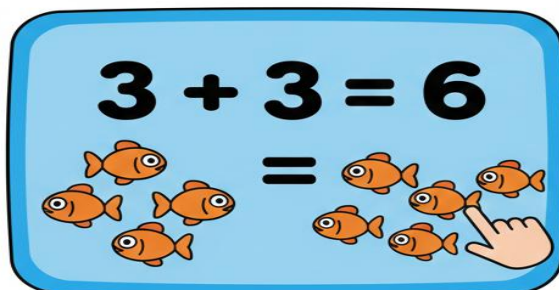
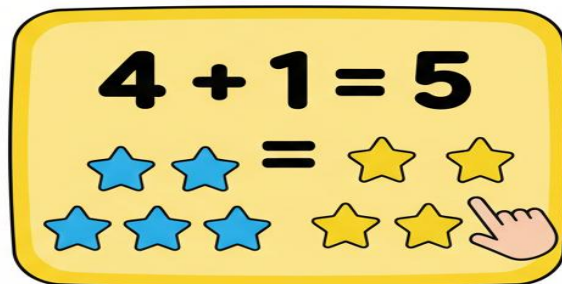
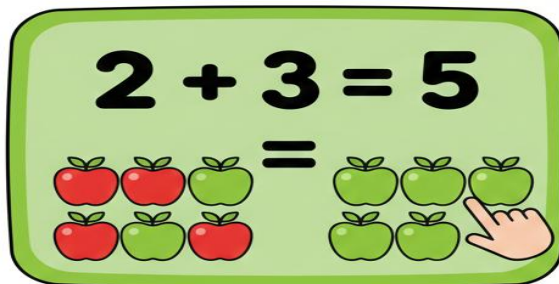
OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Add two numbers with a sum up to 10.
2. State the rule of adding zero to a number.
3. Solve simple word problems involving addition up to 10.
4. Use counters (stones, bottle tops) to solve addition sums.

CONTENT

Adding With Objects!



1. Addition Within 10

- Now we are adding bigger numbers. Using **counters** (stones, buttons, bottle tops) is very helpful.

- **Step-by-step with counters for $5 + 3$:**
 - a. Make a group of 5 counters.
 - b. Make a separate group of 3 counters.
 - c. Push the two groups together to make one big group.
 - d. Count all the counters in the big group: "1, 2, 3, 4, 5, 6, 7, 8."
 - e. The total is **8**. So, **$5 + 3 = 8$** .
- **Using a Number Line:** Start at 5, hop forward 3 spaces. You land on 8.

2. The Rule of Adding Zero

- **Zero is the identity element for addition.** This means adding zero does not change the number.
- **Rule: Any number plus zero equals that same number.**
- **Examples:**
 - If you have 4 toys and get 0 new toys, you still have 4 toys. **$4 + 0 = 4$** .
 - **$5 + 0 = 5$**
 - **$9 + 0 = 9$**

3. Word Problems Within 10

- Read the problem carefully. Identify what is being combined.
- **Example:** "Ada has 6 yams in a basket. Her mom puts 2 more yams in the basket. How many yams are in the basket now?"
 - Numbers: 6 and 2.
 - Operation: Adding (putting more in).
 - Solution: $6 + 2 = 8$ yams.

4. Practical Activities and Strategies

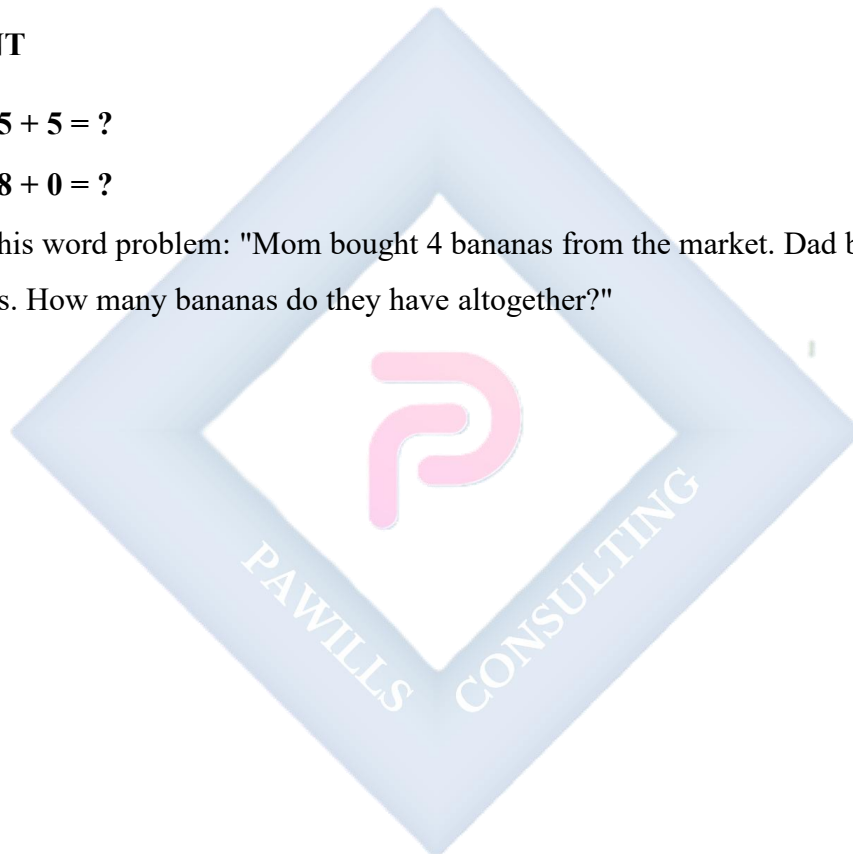
- **Using Fingers:** For $6 + 4$, hold up 6 fingers on one hand and 4 on the other. Count all fingers: 10.
- **Making Ten:** Notice that 6 needs 4 more to make 10. This is a useful strategy for future math.

EVALUATION

1. What is $5 + 2$?
2. What is $7 + 0$?
3. Add these numbers: $4 + 4$.
4. Solve this problem: "If one hen lays 3 eggs and another hen lays 3 eggs, how many eggs are there in total?"
5. Use your counters (e.g., seeds) to find the answer to $6 + 4$.

ASSIGNMENT

1. Solve: $5 + 5 = ?$
2. Solve: $8 + 0 = ?$
3. Solve this word problem: "Mom bought 4 bananas from the market. Dad bought 3 bananas. How many bananas do they have altogether?"



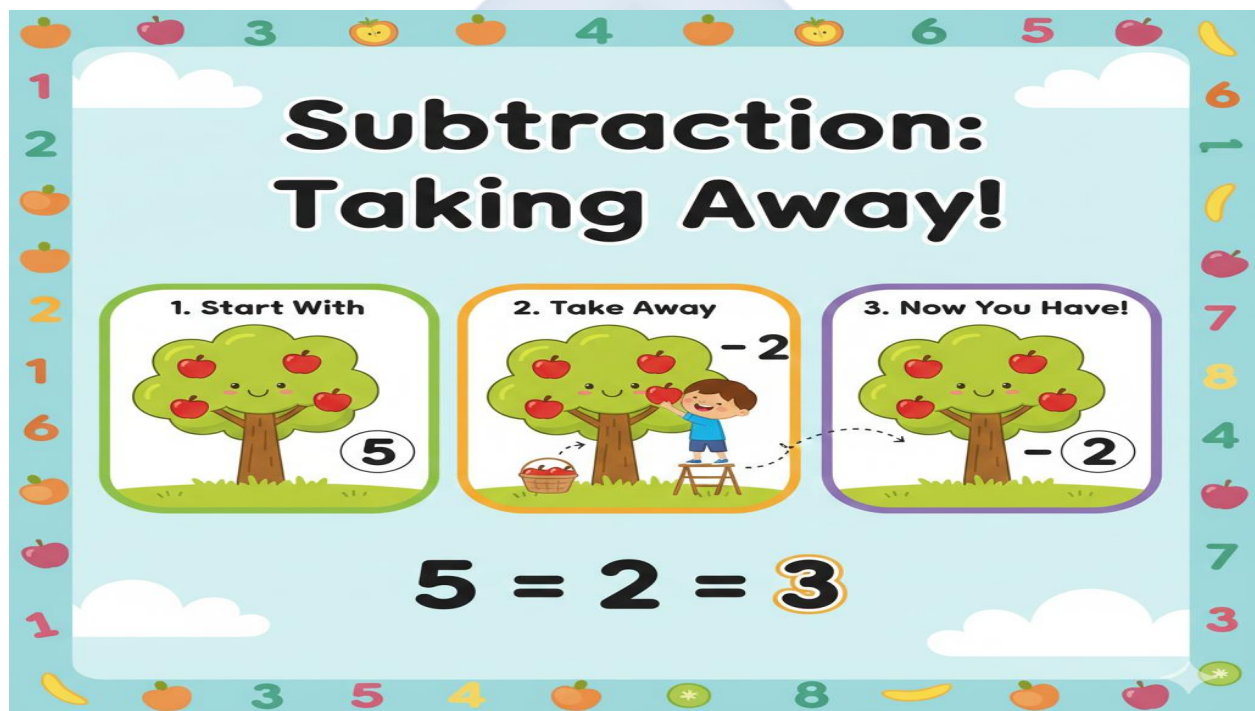
WEEK 9: SUBTRACTION (INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define subtraction as "taking away" or "removing."
2. Identify the minus sign (-).
3. Subtract numbers within 5.
4. Solve simple word problems involving subtraction up to 5.

CONTENT



1. Understanding Subtraction

- **Subtraction** is the process of taking away objects from a group to find out how many are left or remain.
- We use words like **take away**, **minus**, **subtract**, **less**, and **remain**.
- **Concrete Example:** "You have 3 biscuits. If you eat 1 biscuit, how many biscuits are left?"

2. The Sign for Subtraction

- **The Minus Sign (–):** This is a short, straight dash. It tells us to take away or remove one amount from another.

Example: $3 - 1 = 2$

| | |

Starting Take away How many

Amount are left?

3. Simple Subtraction Within 5

- Use physical objects or fingers to model subtraction.
- **Process:** Start with the first number, then **remove** or **count back** the second number.
 - **3 – 1:** Start with 3 fingers. Put 1 finger down. Count the fingers still up: 2. So, $3 - 1 = 2$.
 - **4 – 2:** Start with 4 counters. Take away 2 counters. Count the counters left: 2. So, $4 - 2 = 2$.
- **Subtraction as Counting Backwards:** For $5 - 1$, start at 5 and count back one number: 4.

4. Word Problems Involving Subtraction

- Listen for words like "ate," "gave away," "lost," "fly away," "break" – these often mean subtraction.
- **Example:** "There are 5 mangoes on a plate. We eat 2 mangoes. How many mangoes remain on the plate?"
 - Find the numbers: Start with 5, take away 2.
 - Solve: $5 - 2 = 3$. There are 3 mangoes left.

EVALUATION

1. What does **subtraction** mean?
2. Draw the **minus sign**.
3. What is **3 – 1**?

4. If you have 4 sweets and you eat 1, how many sweets **remain**?
5. Solve: $2 - 2$. (If you take away everything, you have zero left).

ASSIGNMENT

1. Solve this: $5 - 2 = ?$
2. Solve this: $3 - 1 = ?$
3. Draw 4 cups. Cross out (or color over) 2 of them. Write the subtraction sentence to show how many are left (e.g., $4 - 2 = 2$).



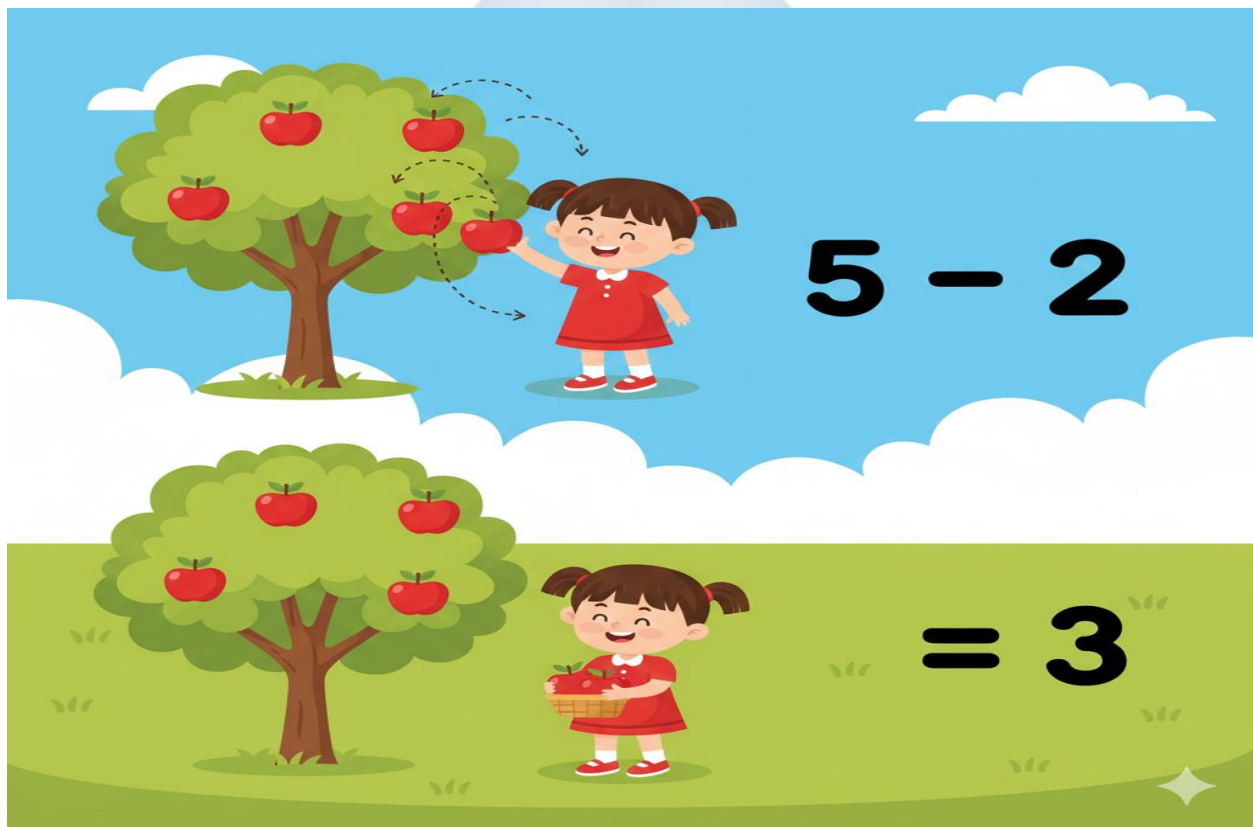
WEEK 10: SUBTRACTION (CONTINUED)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Subtract numbers within 10 using objects.
2. State the result of subtracting zero from a number.
3. Solve word problems involving subtraction up to 10.
4. Demonstrate subtraction using real objects.

CONTENT



1. Subtraction Within 10

- For larger numbers, using counters is essential to see the "taking away" process clearly.
- **Step-by-step with counters for $8 - 3$:**
 - a. Count out 8 counters (this is our whole group).

- b. Physically remove 3 counters from the group and set them aside.
- c. Count the counters that are **left** in the original group: "1, 2, 3, 4, 5."
- d. The number left is **5**. So, $8 - 3 = 5$.
- **Using a Number Line:** Start at 8, hop backwards 3 spaces. You land on 5.

2. The Rule of Subtracting Zero

- If you take away nothing, you keep everything you had.
- **Rule:** Any number minus zero equals that same number.
- **Examples:**
 - If you have 6 books and you take away 0 books, you still have all 6 books. $6 - 0 = 6$.
 - $7 - 0 = 7$
 - $10 - 0 = 10$

3. Word Problems Within 10

- Identify the starting amount and what is being taken away.
- **Example:** "There are 10 birds sitting on a tree branch. 5 birds fly away. How many birds are left on the branch?"
 - Numbers: Start with 10, take away 5.
 - Solution: $10 - 5 = 5$ birds left.

4. Practical Activities and Linking to Addition

- **Activity with Crayons:** Place 9 crayons on the desk. Ask a friend to take away 4 crayons. Count how many are left on the desk (5).
- **Connection:** Subtraction is the opposite of addition. If $5 + 4 = 9$, then $9 - 4$ should equal 5.

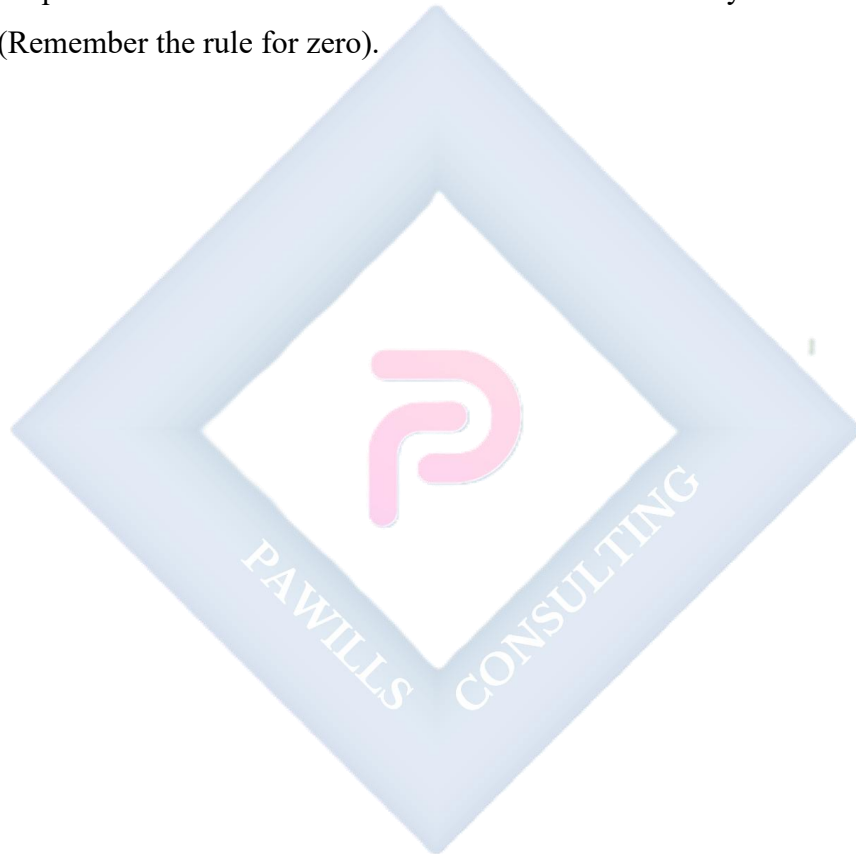
EVALUATION

1. What is $7 - 3$?
2. Solve: $9 - 0$.

3. If you have 10 naira and you spend 5 naira on a sweet, how much money do you have **left**?
4. Take away 2 from 8. (What is $8 - 2$?)
5. Solve: $6 - 6$. (When you take away all you have, you are left with zero).

ASSIGNMENT

1. Solve: $10 - 4 = ?$
2. Solve: $8 - 2 = ?$
3. Solve this problem: "Tolu had 5 balls. He lost 0 balls. How many balls does he have now?" (Remember the rule for zero).



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall numbers 0--10.
2. Solve mixed addition and subtraction problems.
3. Demonstrate readiness for the end-of-term examination.

CONTENT

Review of All Topics

- **Numbers 0-10:**
 - Counting forwards and backwards.
 - Reading and writing numerals.
 - Ordering from smallest to largest.
- **Addition (Putting Together):**
 - Meaning and the plus (+) sign.
 - Adding numbers within 10 (e.g., $4+3=7$).
 - The rule of adding zero.
- **Subtraction (Taking Away):**
 - Meaning and the minus (−) sign.
 - Subtracting numbers within 10 (e.g., $7-2=5$).
 - The rule of subtracting zero.
- **Problem Solving:**
 - Identifying whether a word problem requires addition or subtraction.

EVALUATION

- Oral quiz: Counting from 0 to 10 and back from 10 to 0.
- Board work: Solve simple sums like $3 + 3$ and $5 - 1$.
- Practical: Count a group of objects (up to 10) in the classroom.

ASSIGNMENT

- Study all notes from Week 1 to Week 10.
- Practice writing numbers **0 to 10** neatly in a line.
- Ask a family member to give you two simple addition and two simple subtraction problems to solve.



PRIMARY 1 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Breakdown (Subtopics)
1	Whole Numbers 1–99 (Part 1: 11–20)	- Counting from 11 to 20 - Recognition and identification of numbers 11–20 - Writing numbers 11–20 correctly - Understanding place value: tens and ones (introduction) - Ordering numbers 11–20
2	Whole Numbers 1–99 (Part 2: 21–50)	- Counting from 21 to 50 - Skip counting by 10s (10, 20, 30, 40, 50) - Recognition and writing of numbers 21–50 - Number patterns within 50 - Ordering and comparing numbers within 50
3	Whole Numbers 1–99 (Part 3: 51–99)	- Counting from 51 to 99 - Skip counting by 10s to 90 - Recognition and writing of numbers 51–99 - Place value: tens and ones - Ordering numbers from 1–99 (ascending and descending)
4	Whole Numbers: Quantitative Aptitude on Ordering of Numbers	- Comparing numbers using "greater than" ($>$), "less than" ($<$), and "equal to" ($=$) - Arranging numbers in ascending order (smallest to largest)

		• Arranging numbers in descending order (largest to smallest) • Before, after, and between concepts • Number line activities (0–99)
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday/Rest
7	Whole Numbers: Quantitative Aptitude on Addition and Subtraction	• Addition within 20 (no regrouping) • Subtraction within 20 (no regrouping) • Word problems involving addition and subtraction within 20 • Mental mathematics: quick addition and subtraction • Real-life application of addition and subtraction
8	Open Sentences (1)	• Understanding missing numbers in addition sentences: $3 + \underline{\quad} = 5$ • Finding missing addends using objects • Filling gaps in simple addition number sentences • Using $\square$ or $\underline{\quad}$ to represent missing numbers • Solving simple open sentences within 10
9	Open Sentences (2)	• Understanding missing numbers in subtraction sentences: $5 - \underline{\quad} = 2$ • Finding missing subtrahends using objects • Filling gaps in simple subtraction number sentences

		• Mixed open sentences (addition and subtraction) • Problem-solving with open sentences within 20
10	Money (Introduction)	• Identification of Nigerian coins: ₦1, ₦2 • Identification of Nigerian notes: ₦5, ₦10, ₦20, ₦50 • Understanding the value of different denominations • Counting money using coins and small notes • Simple money transactions: buying and selling with amounts within ₦50
11	REVISION	• Review of all topics: Numbers 1–99, ordering, open sentences, money
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 1 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: WHOLE NUMBERS 1--99 (PART 1: 11--20)

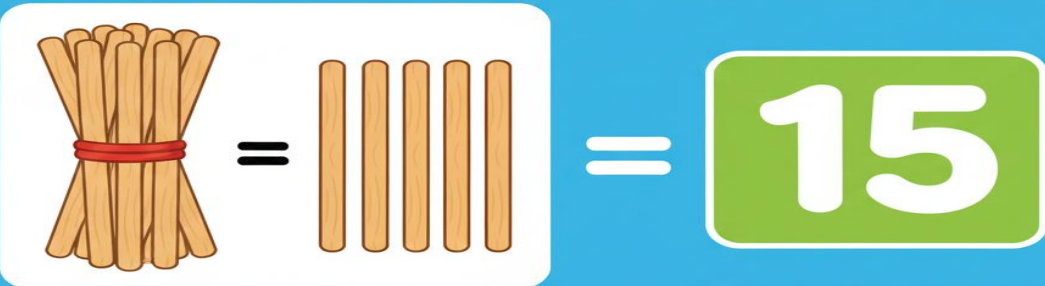
OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count from 11 to 20 correctly.
2. Identify and write the numbers 11 to 20.
3. Explain the concept of place value (Tens and Ones) for numbers 11--19.
4. Arrange numbers 11--20 in the correct order.

CONTENT

**Place Value:
Tens and Ones**



1 Ten 5 Ones = 15

1 Ten + 5 Ones = 15

1. Counting from 11 to 20

- We are moving beyond the "single-digit" numbers (1-9) and the special number 10. Now we enter the "teen" numbers.
- Counting forward from 10: "Ten... what comes next? **Eleven** (11), **Twelve** (12), **Thirteen** (13), **Fourteen** (14), **Fifteen** (15), **Sixteen** (16), **Seventeen** (17), **Eighteen** (18), **Nineteen** (19), **Twenty** (20)."
- **Practical Counting:** Use a pile of 20 counters. Count them aloud as you move them from one pile to another.

2. Recognition and Writing Numbers 11--20

- These numbers are written with two digits side-by-side.
- **Key Pattern:** Notice that numbers 11-19 start with the digit **1**. The second digit tells us the "ones."
- **Number Names and Digits:**
 - **11:** One and One (shows as 1 and 1).
 - **12:** One and Two.
 - **13:** One and Three.
 - **15:** One and Five.
 - **20:** Two and Zero (This is special; it's two full groups of ten).

3. Understanding Place Value: Tens and Ones

- **Place Value** tells us what a digit is worth based on its position in a number.
- The best way to understand this is with "**Bundles of Ten.**"
- **For the number 11:**
 - Make **1** bundle of ten sticks (tied together).
 - Have **1** single stick left over.
 - We say: **1 Ten and 1 One**. This is written as **11**.

Visual for 11:

Tens Place: [1 Bundle] |

Ones Place: [1 Stick] |

Written as: 1 1

- **For the number 17:** It means **1 Ten and 7 Ones**.
- **For the number 20:** It means **2 Tens and 0 Ones** (Two full bundles, with no singles left over).

4. Ordering Numbers 11--20

- To order numbers, we look first at the **Tens** place. All numbers from 11-19 have **1 Ten**. So, we then look at the **Ones** place to decide which is bigger.
- **Example:** 12 vs. 15. Both have 1 Ten. 2 Ones is less than 5 Ones, so 12 is smaller than 15.
- The correct order from smallest to largest is: **11, 12, 13, 14, 15, 16, 17, 18, 19, 20**.

EVALUATION

1. Count aloud from 11 to 15.
2. Write the number **Fifteen**.
3. In the number **14**, how many **tens** are there?
4. In the number **17**, how many **ones** are there?
5. What number comes immediately after 19?
6. Write down the number that means "**1 Ten and 3 Ones**".

ASSIGNMENT

1. Write the numbers from **11 to 20** in your exercise book.
2. Draw **12** balls in your book.
3. Fill in the blank: 1 Ten and 5 Ones = the number ____.

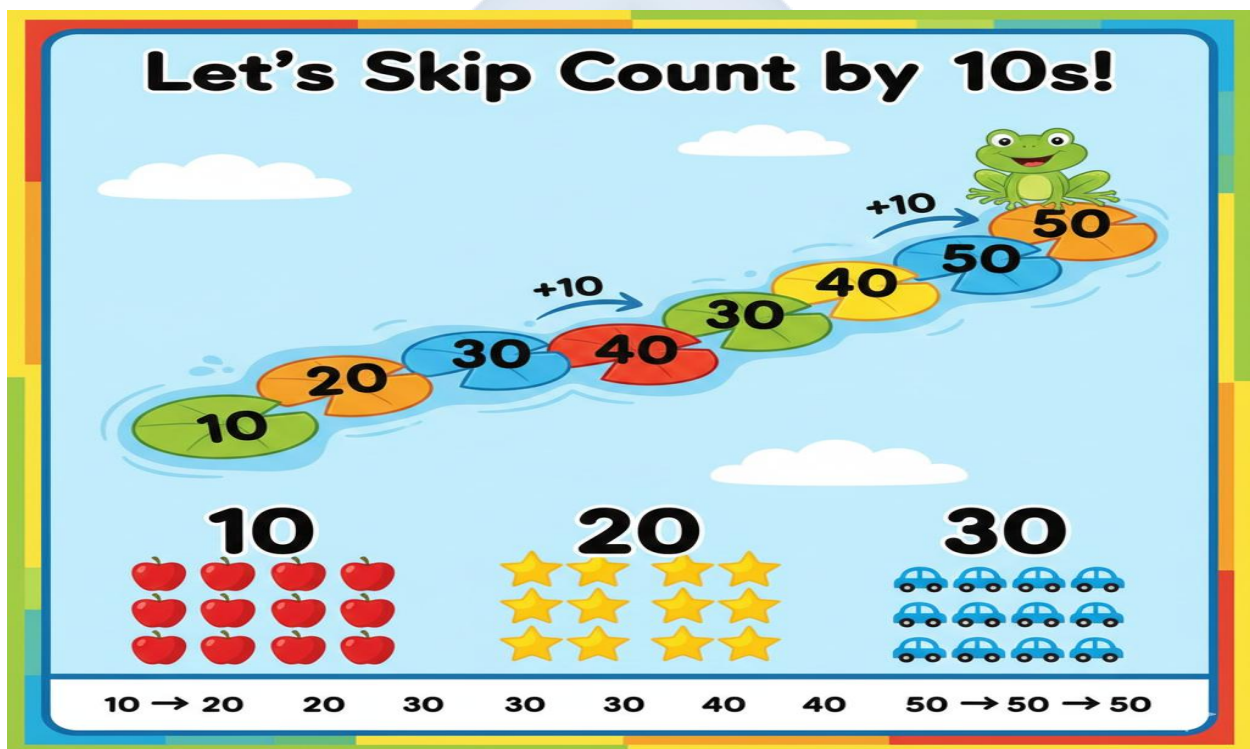
WEEK 2: WHOLE NUMBERS 1--99 (PART 2: 21--50)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count verbally from 21 to 50.
2. Skip count by 10s up to 50.
3. Identify and write numbers from 21 to 50.
4. Order a given set of numbers within 50 from smallest to largest.

CONTENT



1. Counting from 21 to 50

- After 20, we have families of numbers based on the tens.
- **The Twenty-Family:** 21 (Twenty-one), 22 (Twenty-two), 23 ... up to 29 (Twenty-nine).
- **The Thirty-Family:** 30 (Thirty), 31 (Thirty-one), 32 ... up to 39 (Thirty-nine).
- **The Forty-Family:** 40 (Forty), 41 (Forty-one), 42 ... up to 49 (Forty-nine).
- **Fifty:** 50 (Fifty).

2. Skip Counting by 10s

- **Skip counting** is a faster way to count. We jump over numbers, counting only the tens.
- It's like counting bundles: "One bundle of ten (10), two bundles (20), three bundles (30), four bundles (40), five bundles (50)."
- Practice: **10, 20, 30, 40, 50**. This helps us estimate and count large groups quickly.

3. Recognition and Writing of Numbers 21--50

- The first digit tells us the number of **Tens**. The second digit tells us the number of **Ones**.
- **For example:**
 - **34** is "Thirty-four." The **3** means 3 Tens (30). The **4** means 4 Ones. So it's $30 + 4$.
 - **50** is "Fifty." The **5** means 5 Tens (50). The **0** means 0 Ones.

4. Ordering and Comparing Numbers (21-50)

- **Rule 1:** Compare the **Tens** digit first. The number with the larger tens digit is always the bigger number.
 - Example: **42** vs. **28**. 4 Tens is more than 2 Tens, so 42 is bigger than 28.
- **Rule 2:** If the Tens digit is the same, compare the **Ones** digit.
 - Example: **35** vs. **32**. Both have 3 Tens. Compare Ones: 5 is more than 2, so 35 is bigger than 32.

EVALUATION

1. Count aloud from **30 to 40**.
2. Write the number **Forty-five**.
3. Skip count by tens: 10, 20, ____, ____, 50.
4. Which number is bigger: **22** or **42**?
5. What number comes immediately after **49**?
6. Arrange these numbers from smallest to biggest: **35, 20, 50**.

ASSIGNMENT

1. Write the numbers from **21 to 30** clearly in your book.

2. Fill in the missing numbers in this sequence: 31, 32, ____, ____, 35.
3. Circle the biggest number in this list: **29**, **49**, **19**.



WEEK 3: WHOLE NUMBERS 1--99 (PART 3: 51--99)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count from 51 to 99 correctly.
2. Skip count by 10s up to 90.
3. Identify and write numbers 51 to 99.
4. Break down two-digit numbers into Tens and Ones.

CONTENT



1. Counting from 51 to 99

- We continue the pattern we know, now with bigger tens.
- **The Fifty-Family:** 51-59 (Fifty-one to Fifty-nine).
- **The Sixty-Family:** 60-69 (Sixty to Sixty-nine).

- **The Seventy-Family:** 70-79 (Seventy to Seventy-nine).
- **The Eighty-Family:** 80-89 (Eighty to Eighty-nine).
- **The Ninety-Family:** 90-99 (Ninety to Ninety-nine).

2. Skip Counting by 10s to 90

- Let's extend our skip counting song! "Ten, twenty, thirty, forty, fifty, **sixty, seventy, eighty, ninety!**"
- This shows us all the "decades" or groups of ten up to 90.

3. Recognition and Writing

- The sound of the number often hints at the first digit.
 - **Sixty** sounds like **Six** (6).
 - **Eighty** sounds like **Eight** (8).
- **Example:** "Seventy-two" is written as **72** (7 Tens and 2 Ones).

4. Place Value: Tens and Ones (Review and Apply)

- For any two-digit number, the left digit is **Tens** and the right digit is **Ones**.
- **Example for 85:**
 - The digit **8** is in the **Tens** place. It is worth **80** (8 groups of ten).
 - The digit **5** is in the **Ones** place. It is worth **5**.
 - So, **85 = 80 + 5**.
- **Example for 99:** This is the largest two-digit number. It has **9 Tens (90)** and **9 Ones (9)**.
 $90 + 9 = 99$.

5. Ordering Numbers 1--99

- **Ascending Order:** Means from smallest to biggest. (e.g., 1, 10, 50, 99).
- **Descending Order:** Means from biggest to smallest. (e.g., 99, 50, 10, 1).

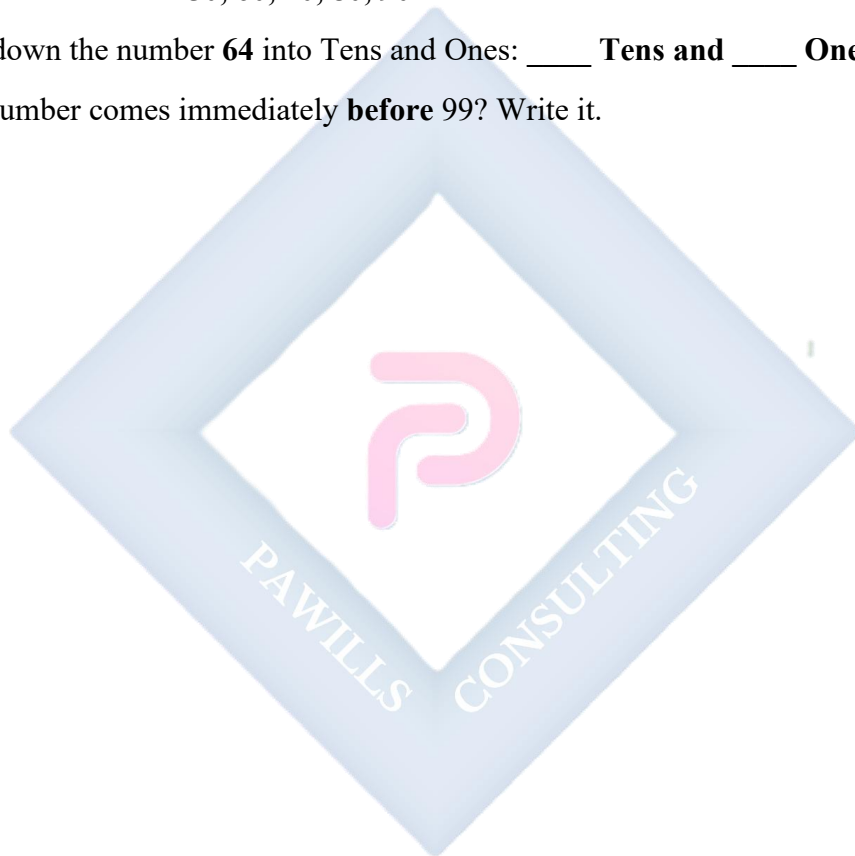
EVALUATION

1. What number comes immediately after **59**?

2. Write the number **Eighty-eight**.
3. Count by tens, starting from 50: 50, ____, ____, ____, 90.
4. In the number **73**, which digit (number) is in the **Tens** place?
5. Arrange these numbers from smallest to biggest: **60, 90, 55**.
6. What number is made of "**9 Tens and 1 One**"?

ASSIGNMENT

1. Write these numbers: **50, 60, 70, 80, 90**.
2. Break down the number **64** into Tens and Ones: ____ **Tens** and ____ **Ones**.
3. What number comes immediately **before** 99? Write it.



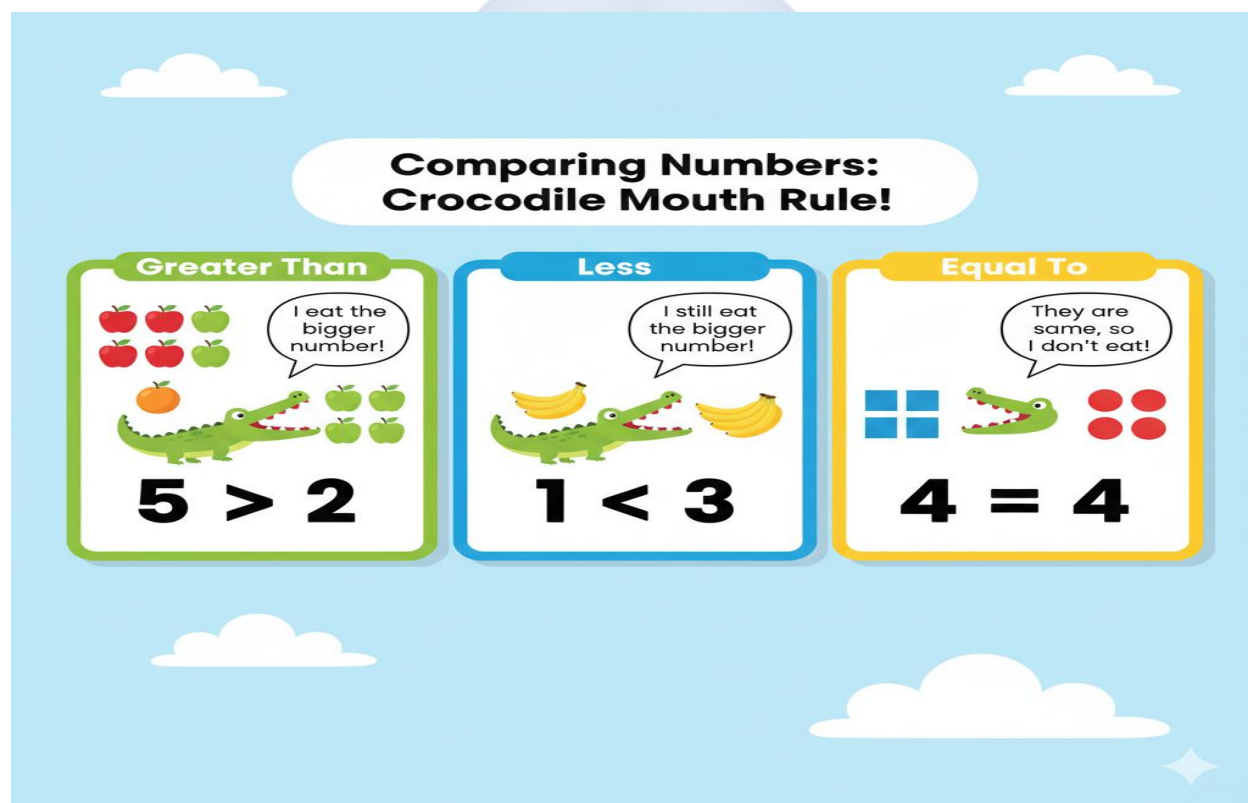
WEEK 4: WHOLE NUMBERS: QUANTITATIVE APTITUDE ON ORDERING

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Compare numbers using the symbols $>$ (greater than), $<$ (less than), and $=$ (equal to).
2. Arrange a set of numbers in ascending order.
3. Arrange a set of numbers in descending order.
4. Identify numbers that come before, after, or between other numbers.

CONTENT



1. Comparing Numbers ($>$, $<$, $=$)

- These symbols are like signs that tell us the relationship between two numbers.
- **The Crocodile Mouth Rule:** Imagine the symbol is a hungry crocodile's mouth. It always wants to eat the **bigger** number!

- **> (Greater Than):** The mouth opens to the left. The bigger number is on the wide side. Example: **10 > 5** (Ten is greater than five).
- **< (Less Than):** The mouth opens to the right. The bigger number is on the wide side. Example: **2 < 8** (Two is less than eight).
- **= (Equal to):** Two lines that are the same length. It means both sides have the same value. Example: **5 = 5**.

2. Ascending Order

- **Ascending** means going **up**, like climbing stairs. We start with the smallest number and go up to the biggest.
- **Example:** Arrange 5, 12, 30, 45 in ascending order: **5, 12, 30, 45**.

3. Descending Order

- **Descending** means going **down**, like walking down stairs. We start with the biggest number and go down to the smallest.
- **Example:** Arrange 45, 30, 12, 5 in descending order: **45, 30, 12, 5**.

4. Before, After, Between

- These words describe a number's position in a sequence.
- **Before:** The number that comes earlier. (The number before 10 is **9**).
- **After:** The number that comes next. (The number after 10 is **11**).
- **Between:** The number in the middle of two others. (The number between 10 and 12 is **11**).

EVALUATION

1. Put the correct sign (> or <) in the blank: 20 _____ 10.
2. Put the correct sign (> or <) in the blank: 5 _____ 9.
3. Arrange in **ascending** order (smallest to biggest): 8, 2, 10.
4. Arrange in **descending** order (biggest to smallest): 15, 50, 5.
5. What number is **between** 18 and 20?

6. Is 40 **equal to** 40? Answer Yes or No.

ASSIGNMENT

1. Use $>$ or $<$: 90 _____ 91.
2. Write the numbers that come **between** 30 and 34.
3. Arrange these numbers from the **biggest to the smallest**: 10, 5, 20, 15.



WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Week 1 to Week 4.

CONTENT

Assessment of Weeks 1–4

- Counting, reading, and writing numbers from **11 to 99**.
- Understanding **Place Value** (Tens and Ones).
- Using comparison symbols: $>$, $<$, $=$.
- **Ordering** numbers (Ascending and Descending).

EVALUATION

- Administer the written Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

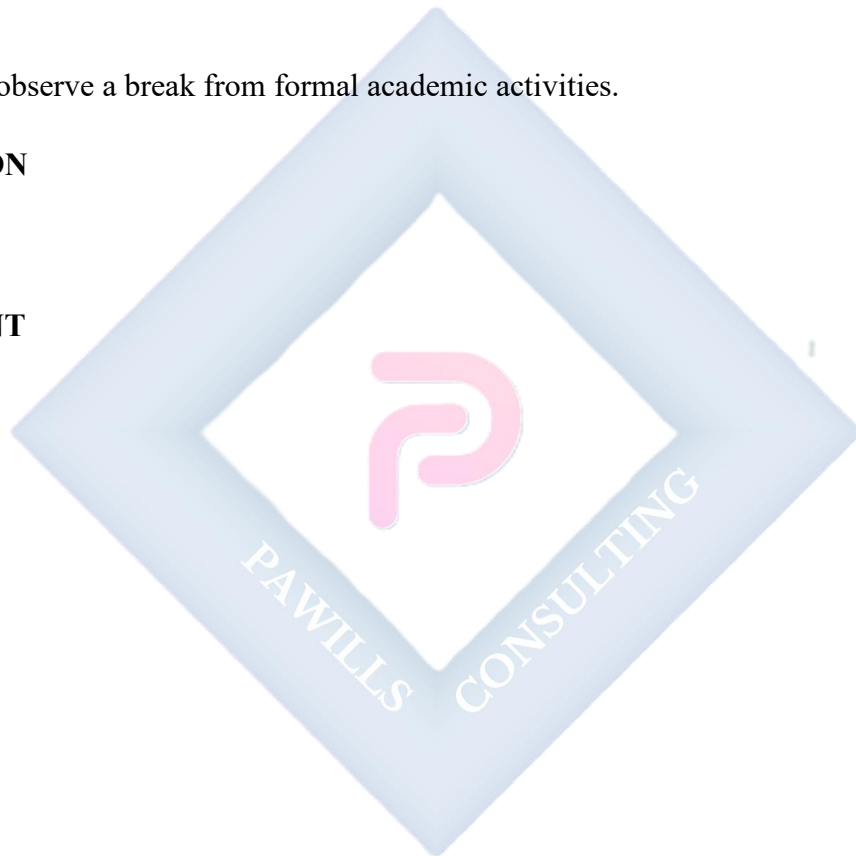
- Pupils observe a break from formal academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



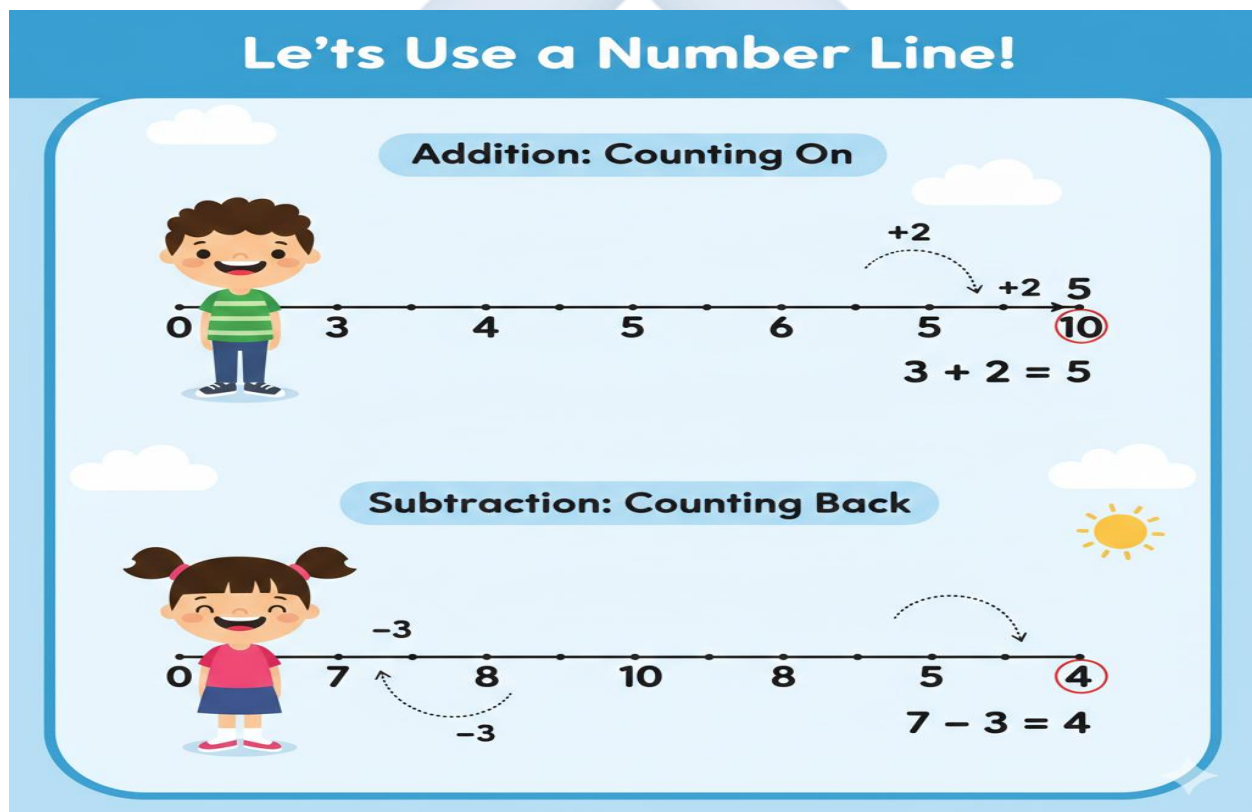
WEEK 7: WHOLE NUMBERS: QUANTITATIVE APTITUDE ON ADDITION AND SUBTRACTION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Add two numbers with a sum up to 20 without regrouping.
2. Subtract a smaller number from a larger number within 20.
3. Solve simple word problems involving addition and subtraction.
4. Perform mental addition of single-digit numbers.

CONTENT



1. Addition within 20 (Without Regrouping)

- We are adding numbers where the Ones add up to 9 or less, so we don't need to make a new Ten.

- **Strategy: Count On.** Start with the bigger number in your head, then count on the smaller number using your fingers.
- **Example: $12 + 3 = ?$**
 - Think: "12... count on 3 more: 13, 14, 15."
 - Answer: **15.**
- **Using a Number Line:** Start at 12, hop forward 3 spaces. You land on 15.

2. Subtraction within 20

- We are taking away from numbers up to 20.
- **Strategy: Count Back.** Start with the bigger number, then count back the smaller number.
- **Example: $15 - 4 = ?$**
 - Think: "15... count back 4: 14, 13, 12, 11."
 - Answer: **11.**
- **Using Objects:** Draw 15 sticks. Cross out 4. Count the remaining sticks.

3. Word Problems (Within 20)

- Read carefully. Decide if the problem is asking for a **total** (add) or what is **left** (subtract).
- **Addition Example:** "Ada has 10 yams. She buys 5 more. How many yams does she have **altogether**?" ($10 + 5 = 15$)
- **Subtraction Example:** "There are 18 birds on a wire. 3 birds fly away. How many birds are **left** on the wire?" ($18 - 3 = 15$)

4. Mental Mathematics (Facts to 10)

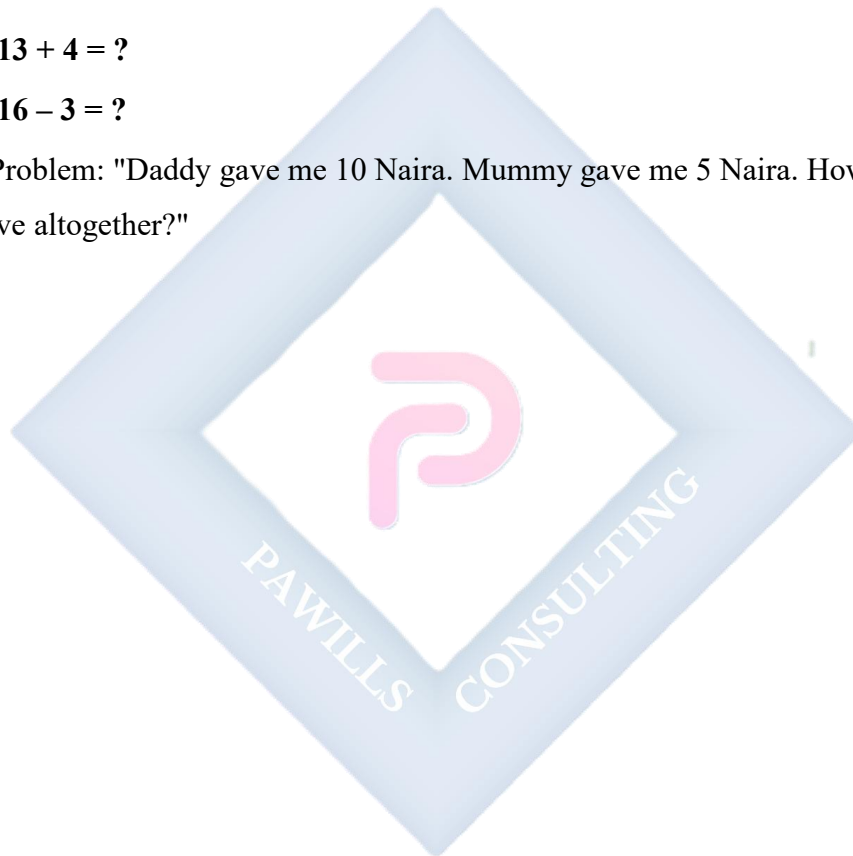
- These are basic number facts we should know by heart for quick calculation.
- **Doubles:** $2 + 2 = 4$, $5 + 5 = 10$.
- **Making Ten:** $7 + 3 = 10$, $8 + 2 = 10$.
- **Subtraction from Ten:** $10 - 1 = 9$, $10 - 4 = 6$.

EVALUATION

1. Add: **11 + 5**.
2. Subtract: **14 – 2**.
3. Word Problem: "Musa has 10 goats. 2 goats run away. How many goats are left?"
4. What is **10 + 10**?
5. Solve: **19 – 5**.
6. Mental Math: If I have 12 sweets and I get 1 more, how many do I have?

ASSIGNMENT

1. Solve: **13 + 4 = ?**
2. Solve: **16 – 3 = ?**
3. Word Problem: "Daddy gave me 10 Naira. Mummy gave me 5 Naira. How much money do I have altogether?"



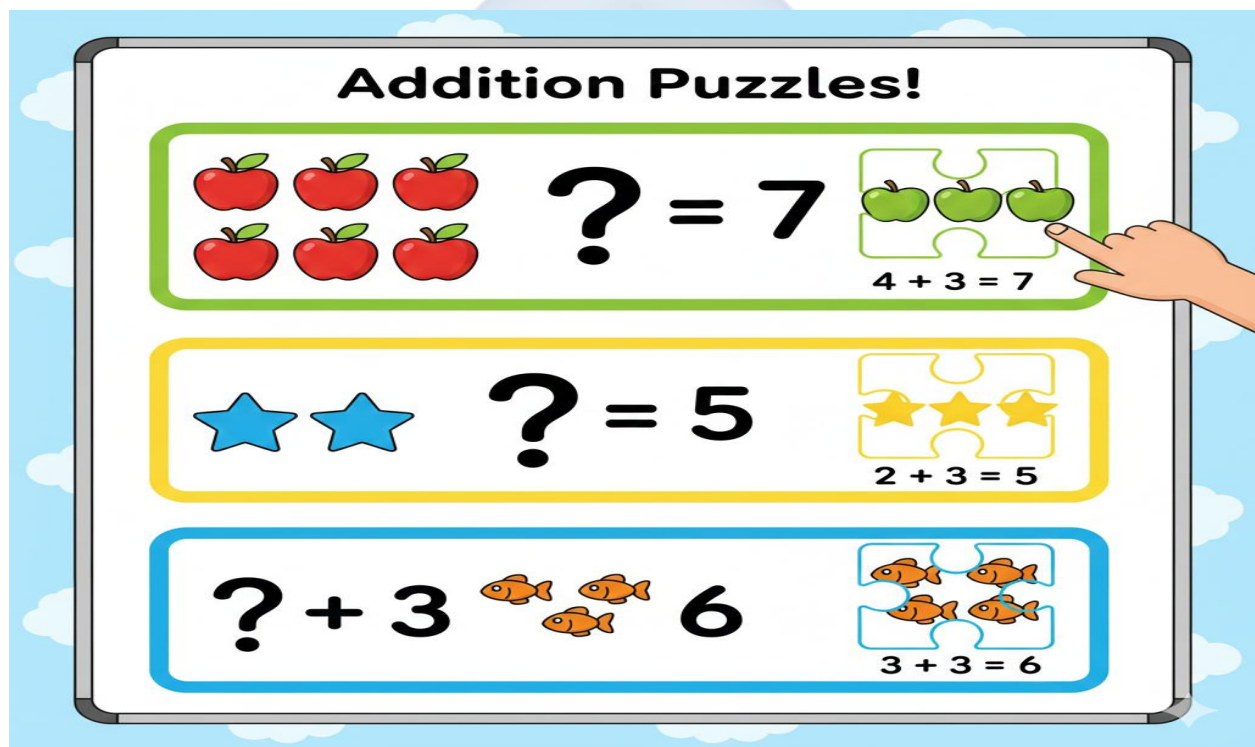
WEEK 8: OPEN SENTENCES (1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the missing number in a simple addition sentence.
2. Use objects to find a missing addend.
3. Solve open sentences where the sum is up to 10.
4. Use a box ($\square$) to represent a missing number.

CONTENT



1. Understanding Missing Numbers (Open Sentences)

- An **open sentence** is like a puzzle with a missing piece. It has a box ($\square$) or a line ($\underline{\quad}$) where a number should be.
- Example: $3 + \square = 5$. This asks: "3 plus *what number* equals 5?"

2. Finding Missing Addends using Objects

- The best way to solve this is with physical objects (counters, stones).
- **To solve $3 + \square = 5$:**
 - a. Take 3 counters. This is what we start with.
 - b. The total should be 5. Keep adding counters one by one to your group of 3 until you have 5 in total.
 - c. You added 2 counters. So, the missing number is 2. The sentence is $3 + 2 = 5$.

3. Using Mental Strategies

- **Think Addition in Reverse (Subtraction):** For $2 + \underline{\quad} = 6$, think " $6 - 2 = \text{what?}$ " The answer is 4.
- **Count On:** For $5 + \square = 9$, start at 5 and count on until you reach 9: "6, 7, 8, 9." That's 4 counts. The missing number is 4.

4. Solving Simple Sentences within 10

- Let's practice with numbers we know well.
- **Example: $5 + \square = 10$.** Think: "What do I add to 5 to make 10?" We know $5 + 5 = 10$. The missing number is 5.
- **Special Case: Adding Zero.** $4 + \square = 4$. What can you add to 4 to keep it 4? Only 0. So, $4 + 0 = 4$.

EVALUATION

1. Find the missing number: $2 + \square = 4$.
2. Complete: $1 + \underline{\quad} = 5$.
3. Solve: $4 + \square = 4$.
4. $\underline{\quad} + 3 = 6$.
5. Use your fingers to solve: $5 + \square = 7$.

ASSIGNMENT

1. Copy and complete: $3 + \square = 8$.
2. Copy and complete: $6 + \square = 10$.

3. Draw circles to show this problem: Start with 2 circles. Draw more circles until you have 5 in total. Write the open sentence: $2 + \square = 5$.



WEEK 9: OPEN SENTENCES (2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the missing number in a simple subtraction sentence.
2. Find missing subtrahends using counters.
3. Solve mixed open sentences (addition and subtraction) within 20.
4. Determine the missing number when given the answer.

CONTENT

Open Sentences: Subtraction Fun!

The Puzzle


$$7 - \boxed{?} = 2$$

The Answer


$$7 - \boxed{5} = 2$$

We took away 5 apples!

1. Understanding Missing Numbers in Subtraction

- A subtraction open sentence also has a missing part.
- Example: $5 - \square = 2$. This asks: "5 take away *what number* leaves 2?"

2. Finding Missing Subtrahends

- Use counters to solve.

- **To solve $5 - \square = 2$:**
 - a. Start with 5 counters (the total we begin with).
 - b. We know we end up with 2 counters left.
 - c. Remove counters one by one from the 5 until only 2 are left.
 - d. You removed **3** counters. So, the missing number is **3**. The sentence is **$5 - 3 = 2$** .

3. Filling Gaps in Sentences

- **Strategy: Think Addition.** If $10 - \square = 5$, think "5 plus what equals 10?" $5 + 5 = 10$, so the missing number is **5**.
- **Strategy: Count Down.** For $15 - \square = 10$, start at 15 and count down until you reach 10: "14, 13, 12, 11, 10." That's 5 counts down. The missing number is **5**.

4. Mixed Problems within 20

- Now we must read carefully to see if the open sentence is addition or subtraction.
- **Example 1 (Addition):** $\square + 2 = 12$. Think "What plus 2 equals 12?" $10 + 2 = 12$.
Answer: **10**.
- **Example 2 (Subtraction):** $15 - \square = 14$. Think "15 take away what is 14?" $15 - 1 = 14$.
Answer: **1**.

EVALUATION

1. Find the missing number: $4 - \square = 1$.
2. Complete: $10 - \square = 8$.
3. Solve: $5 - \square = 0$. (If you take away everything, you have zero left).
4. $\square + 2 = 12$.
5. $15 - \square = 14$.
6. Word Problem: "If I have 6 eggs and I break some, now I have 4. How many did I break?" (Write the open sentence: $6 - \square = 4$).

ASSIGNMENT

1. Copy and complete: $8 - \square = 4$.

2. Copy and complete: $20 - \square = 10$.

3. Copy and complete: $7 + \square = 9$.



WEEK 10: MONEY (INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify Nigerian coins (Kobo and Naira).
2. Identify Nigerian bank notes (₦5, ₦10, ₦20, ₦50).
3. State the colour of the ₦10, ₦20, and ₦50 notes.
4. Perform simple buying and selling sums within ₦50.

CONTENT



1. Identification of Nigerian Coins

- **Money** is what we use to pay for goods and services. Nigeria's currency is the **Naira and Kobo**.
- **Coins** are the hard, round money made from metal.
- Common coins include **50 Kobo**, **₦1**, and **₦2**. (Note: Show real or realistic pictures. Explain that coins are used for smaller amounts).

2. Identification of Nigerian Notes

- **Notes** are the paper money. They have different values, colours, and pictures.
- **₦5 Note (Five Naira):** Colour is cream or brownish.
- **₦10 Note (Ten Naira):** Colour is red or orange.
- **₦20 Note (Twenty Naira):** Colour is green.
- **₦50 Note (Fifty Naira):** Colour is blue.
- **Security Tip:** Notes have special marks (like raised print) to help tell if they are real.

3. Comparing the Value of Money

- Not all naira are equal! A ₦50 note can buy more than a ₦5 note because it has a higher **value**.
- Value Order (Smallest to Largest): $₦5 < ₦10 < ₦20 < ₦50$.

4. Simple Transactions (Buying and Selling)

- This uses the addition and subtraction skills we have learned.
- **Addition for Cost:** If a sweet costs ₦5 and a biscuit costs ₦10, buying both costs $₦5 + ₦10 = ₦15$.
- **Subtraction for Change:** If you give the seller ₦20 for an item that costs ₦10, your change is $₦20 - ₦10 = ₦10$.

EVALUATION

1. What is the name of the money we use in Nigeria?
2. What colour is the **₦20** note?
3. Which is worth more money: a **₦50** note or a **₦5** note?
4. If one pencil costs **₦10**, how much will two pencils cost?
5. Which Nigerian note is blue in colour?

ASSIGNMENT

1. Draw a rectangle and colour it **green**. Write **₦20** inside it to show a Twenty Naira note.

2. Ask your parents to show you a **₦50** note. Look at it carefully. Write down one thing you see on it (e.g., a picture, numbers).
3. Solve this money problem: **₦10 + ₦10 = ?**



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from numbers 1--99, open sentences, and money.
2. Demonstrate readiness for the end-of-term examination.

CONTENT

Review of All Topics

- **Numbers 11-99:**
 - Counting in tens and ones.
 - Place Value (Understanding Tens and Ones in numbers like 36).
 - Comparing ($>$, $<$, $=$) and ordering numbers.
- **Calculations:**
 - Addition and subtraction within 20.
 - Solving **Open Sentences** (finding missing numbers in addition and subtraction puzzles).
- **Money:**
 - Identifying Nigerian notes (₦5, ₦10, ₦20, ₦50) by colour and value.
 - Simple addition and subtraction with naira.

EVALUATION

- Class quiz: Quick identification of numbers and note values.
- Board work: Solving open sentences like $8 + \square = 15$ and $12 - \square = 7$.
- Oral recitation: Skip counting by tens up to 90.

ASSIGNMENT

- Study all notes from Week 1 to Week 10 in preparation for the end-of-term examination.

PRIMARY 1 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Breakdown (Subtopics)
1	Fractions (Introduction)	• Understanding the concept of "half" ($\frac{1}{2}$) • Dividing objects and shapes into two equal parts • Identifying and shading half of shapes (circles, rectangles, squares) • Practical activities: sharing items equally between two people • Introduction to "quarter" ($\frac{1}{4}$) using folding and cutting
2	Length (Measurement)	• Understanding long and short • Comparing lengths: longer, shorter, longest, shortest • Measuring length using non-standard units (hand span, foot, pencil, stick) • Arranging objects by length • Introduction to standard units: metre and centimetre (basic awareness)
3	Weight	• Understanding heavy and light • Comparing weights: heavier, lighter, heaviest, lightest • Using balance scales to compare weights • Weighing objects using non-standard units (stones, blocks) • Practical activities: comparing weights of classroom objects
4	Time	• Understanding concepts: day, night, morning, afternoon, evening

		• Days of the week (Monday to Sunday) • Understanding yesterday, today, tomorrow • Introduction to clock: identifying the clock face • Reading time: o'clock times only (1 o'clock, 2 o'clock, etc.)
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday/Rest
7	Three Dimensional Shapes	• Identification of 3D shapes: cube, cuboid, sphere, cylinder, cone • Properties of 3D shapes: faces, edges, corners • Sorting and classifying 3D shapes • Finding 3D shapes in the environment (ball, box, tin, etc.) • Building with 3D shapes
8	Two Dimensional Shapes	• Identification of 2D shapes: circle, triangle, square, rectangle, oval • Properties of 2D shapes: sides and corners • Drawing basic 2D shapes • Sorting and classifying 2D shapes • Finding 2D shapes in the environment and on objects
9	Two Dimensional Shapes (Continued) & Patterns	• Creating patterns with 2D shapes • Relationship between 2D and 3D shapes (faces of cubes are squares, etc.) • Combining shapes to make pictures • Symmetry: understanding same on both sides • Practical art activities with shapes
10	Data Collection (Introduction)	• Understanding data collection: gathering

		information • Collecting simple data about the class (favourite fruits, colours, pets) • Using tally marks to count and record • Organizing data in simple tables • Reading and interpreting simple pictographs (one picture = one object) • Creating simple bar charts using blocks or drawings
11	REVISION	• Review of all topics: Fractions, measurement (length, weight, time), shapes, data collection
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 1 MATHEMATICS LESSON NOTES

THIRD TERM

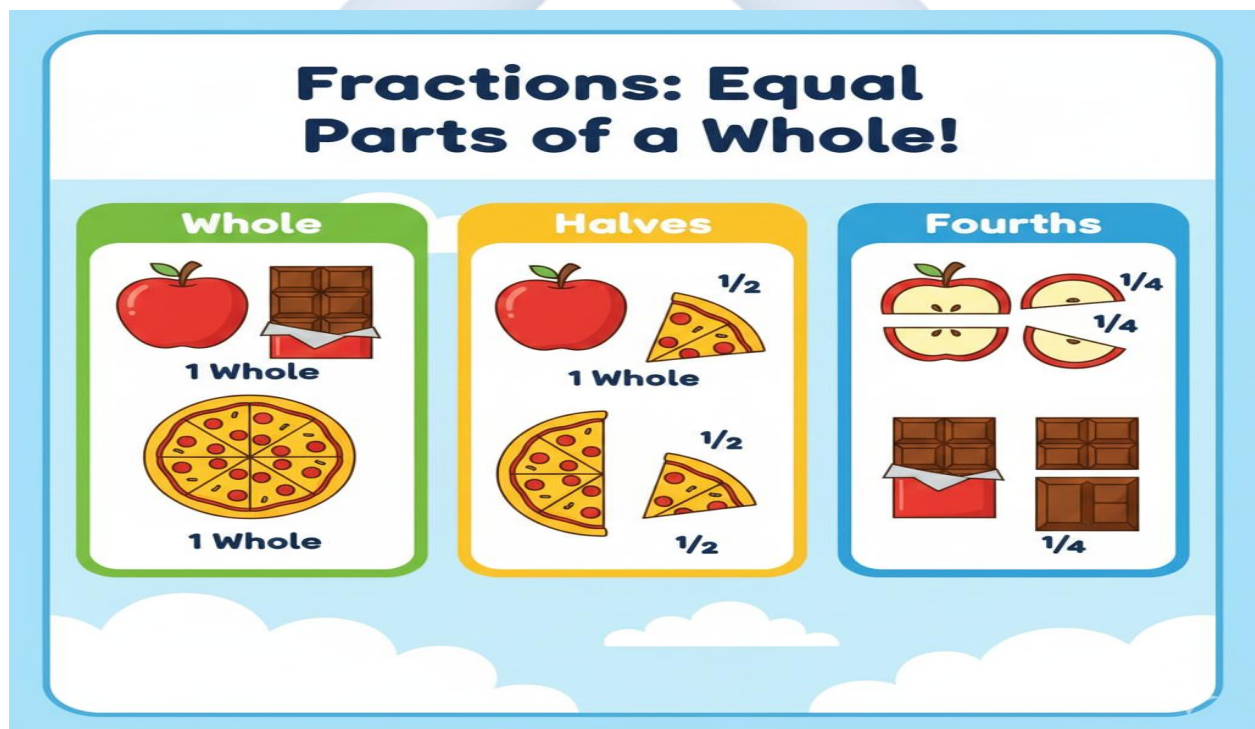
WEEK 1: FRACTIONS (INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define a fraction as a part of a whole.
2. Identify "half" ($\frac{1}{2}$) as one of two equal parts.
3. Identify "quarter" ($\frac{1}{4}$) as one of four equal parts.
4. Shade half of a given shape correctly.

CONTENT



1. Understanding the Concept of "Half"

- A **fraction** means a part or a piece of something whole. We use fractions when we share or divide things.

- A **whole** is one complete object, like a full orange, a whole bar of chocolate, or a full cup of water.
- When we divide one whole thing into **two equal parts** (two parts that are exactly the same size), each part is called a **half**.
- We write half with a special symbol: $\frac{1}{2}$. The number 1 on top shows we have 1 part. The number 2 at the bottom shows the whole was cut into 2 equal parts.
- **The Key is Equal:** If one piece is big and the other is small, they are **not** halves.

2. Identifying and Shading Half

- We can find half of a shape by folding or drawing a line down the middle so that both sides match perfectly.
- **Activity:** Give pupils a paper rectangle. Ask them to fold it once so that both sides are equal. The fold line creates two halves.
- **Shading:** To show one half, we shade one of the two equal parts.

Example: A rectangle divided into halves.

+-----+	+-----+
XXXXXXXXX	
XXXXXXXXX	or
XXXXXXXXX	
+-----+	+-----+
Shaded Half	Unshaded Half

3. Introduction to "Quarter"

- When we divide one whole into **four equal parts**, each part is called a **quarter** or a **fourth**.
- We write a quarter as $\frac{1}{4}$. The 1 on top means we have 1 part. The 4 at the bottom means the whole was cut into 4 equal parts.
- **Making Quarters:** You can make quarters by folding a paper in half, and then in half again. You will see 4 equal sections.

4. Sharing Equally as Fractions

- Fractions are fair. They help us share things equally.
- **Example of Half:** If you have one apple and you want to share it equally with **one friend**, you cut it into **two halves**. You get $\frac{1}{2}$, your friend gets $\frac{1}{2}$.
- **Example of Quarter:** If you have one apple and want to share it equally with **three friends**, you cut it into **four quarters**. Each person gets $\frac{1}{4}$.

EVALUATION

1. What do we call one part when we divide an object into two equal parts?
2. Write the symbol for half.
3. If I cut an orange into 4 equal slices, what is one slice called?
4. Is a small piece of bread and a big piece of bread "equal halves"? Why or why not?
5. How many halves are needed to make one whole?

ASSIGNMENT

1. Draw a square. Divide it into two equal parts with a line and shade one half ($\frac{1}{2}$) of it.
2. Draw a circle and divide it into four equal quarters (4 parts).
3. At home, fold a piece of paper into two equal parts. Paste it in your book and write " $\frac{1}{2}$ " next to it.

WEEK 2: LENGTH (MEASUREMENT)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Differentiate between "long" and "short" objects.
2. Compare two objects using the words "longer" and "shorter".
3. Measure an object using non-standard units (hand span or pencil).
4. Arrange three objects from shortest to longest.

CONTENT



1. Understanding Long and Short

- **Length** tells us how long something is from one end to the other. It is a measure of distance.

- **Long:** Something that has a great distance from end to end. Example: A snake, a ruler, a skipping rope.
- **Short:** Something that has a small distance from end to end. Example: A crayon, a key, your finger.
- The same object can be long or short compared to different things. A pencil is long compared to an eraser but short compared to a broom.

2. Comparing Lengths

- We compare by placing two objects side-by-side, starting at the same point.
- The object that **extends further** is **longer**.
- The object that **does not reach as far** is **shorter**.
- **Longest/Shortest:** When comparing three or more objects, the one that extends the most is the **longest**. The one that extends the least is the **shortest**.

3. Measuring with Non-Standard Units

- Before we use rulers, we can measure using body parts or everyday objects. This is called **non-standard measurement**.
- **Hand Span:** The distance from the tip of your thumb to the tip of your little finger when your hand is stretched out.
- **Foot:** The length of your foot.
- **Pencil/Stick:** Using the same pencil to see how many times it fits along an object.
- **Example:** "My table is 6 hand spans long."

4. Introduction to Standard Units

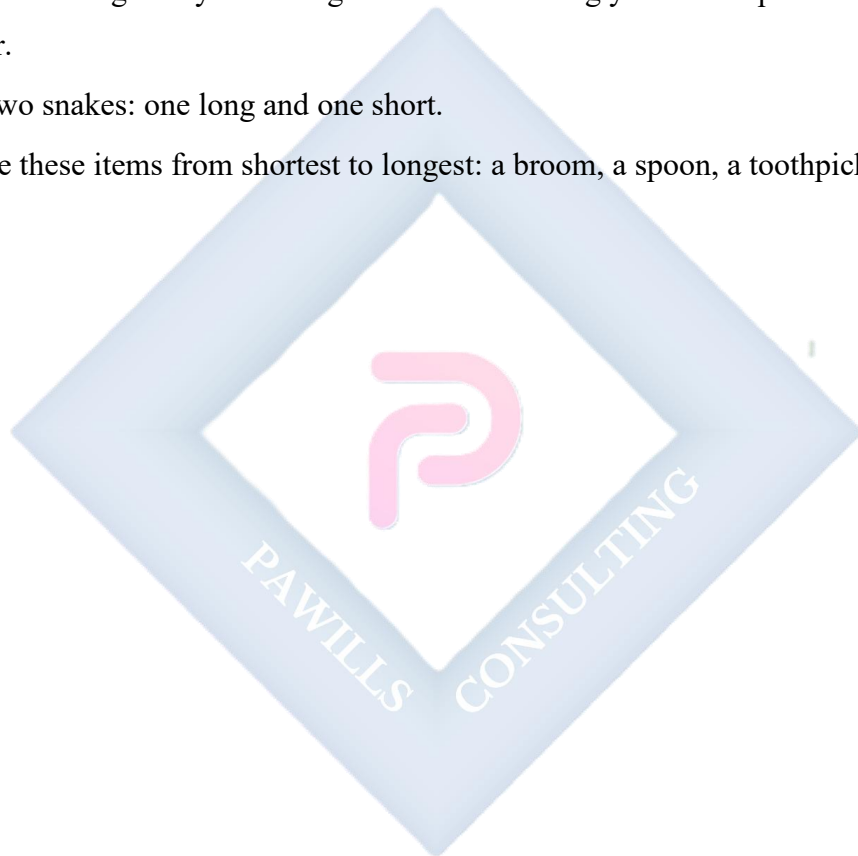
- Why do we need rulers? Because my hand span and your hand span are different sizes! To give the same answer everywhere, we use **standard units**.
- **Centimetres (cm)** and **Metres (m)** are standard units. A ruler has centimetres marked on it.

EVALUATION

1. Which is usually longer: a new pencil or an eraser?
2. Use your hand span to measure the width of your desk. How many spans is it?
3. Who is taller: You or your teacher? Who is shorter?
4. What is the name of the measurement we do using our spread fingers?
5. Name one object in this classroom that is short.

ASSIGNMENT

1. Measure the length of your dining table at home using your hand span. Write down the number.
2. Draw two snakes: one long and one short.
3. Arrange these items from shortest to longest: a broom, a spoon, a toothpick.



WEEK 3: WEIGHT

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Differentiate between "heavy" and "light" objects.
2. Compare two objects using "heavier" and "lighter".
3. Explain how a balance scale works.
4. Estimate the weight of common classroom objects.

CONTENT



1. Understanding Heavy and Light

- **Weight** tells us how heavy something is. It is the measure of how much gravity pulls on an object.

- **Heavy:** An object that is hard to lift or pick up. It feels like it is being pulled down strongly. Examples: A big rock, a filled school bag, a refrigerator.
- **Light:** An object that is easy to lift or pick up. It feels like it has very little pull. Examples: A feather, a sheet of paper, a balloon.
- Size doesn't always mean heavy! A big balloon is light. A small stone is heavy.

2. Comparing Weights

- The best way to compare is to hold one object in each hand.
- The hand that gets pulled down more holds the **heavier** object.
- The hand that is lifted up more easily holds the **lighter** object.
- **Heaviest/Lightest:** When comparing many objects, the one that is hardest to lift is the **heaviest**. The easiest to lift is the **lightest**.

3. Using Balance Scales

- A **balance scale** is a tool that helps us compare weights very clearly.
- It has two pans that hang from a central point.
- **Rule:** The pan with the **heavier** object will go **DOWN**. The pan with the **lighter** object will go **UP**.
- If both pans are level, the weights are **equal** or balanced.

Balanced	Left is Heavier
/ \	/ \
/ \	/ \
[] []	[] []
Same weight	Down Up

4. Weighing with Non-Standard Units

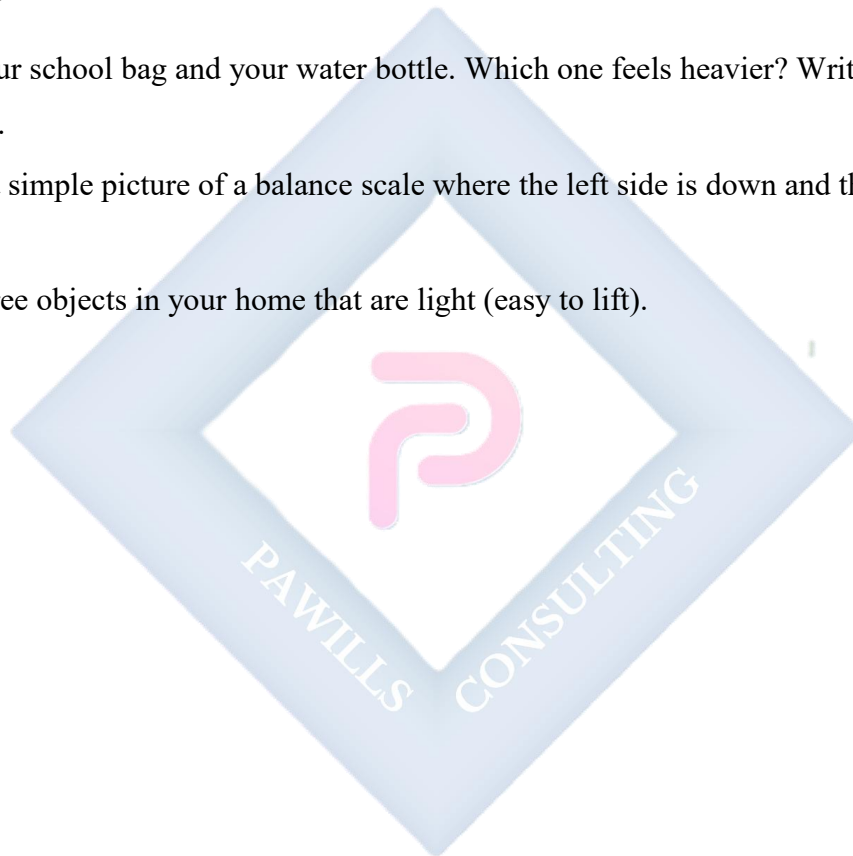
- We can use identical objects (like blocks, stones, or beans) to measure weight.
- **Example:** "The storybook weighs the same as 12 wooden blocks."

EVALUATION

1. Which is heavier: a teacher's table or a pupil's chair?
2. Which is lighter: a feather or a stone?
3. On a balance scale, does the heavy object make the pan go up or down?
4. Name one heavy thing you can find in your house.
5. Compare a shoe and a sock by feeling them. Which one is lighter?

ASSIGNMENT

1. Lift your school bag and your water bottle. Which one feels heavier? Write down your answer.
2. Draw a simple picture of a balance scale where the left side is down and the right side is up.
3. List three objects in your home that are light (easy to lift).



WEEK 4: TIME

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the parts of the day (morning, afternoon, evening, night).
2. List the days of the week in the correct order.
3. Identify the face of a clock and its hands.
4. Read the time when the long hand is on 12 (O'clock).

CONTENT



1. Concepts of Day and Night

- Time helps us organize our day. One full day includes daytime and nighttime.
- **Morning:** From sunrise until about 12 noon. The sun is rising. We wake up, brush our teeth, and go to school.

- **Afternoon:** From about 12 noon until 4 or 5 p.m. The sun is high. We eat lunch and have afternoon lessons.
- **Evening:** From about 5 p.m. until the sun sets and it gets dark. We eat dinner, do homework, and relax with family.
- **Night:** When it is dark outside. The moon is out. We sleep.

2. Days of the Week

- There are **7 days** in one week. Each day has a name.
- The correct order is: **Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.**
- **Yesterday:** The day that just passed.
- **Today:** The day we are in now.
- **Tomorrow:** The day that is coming next.

3. Introduction to the Clock (Analog)

- A clock is a tool that tells us the exact time.
- **Clock Face:** The round part with numbers from **1 to 12**.
- **Clock Hands:** The pointers that move.
 - **Short Hand (Hour Hand):** This hand is shorter and thicker. It tells us the **hour**.
 - **Long Hand (Minute Hand):** This hand is longer and thinner. It tells us the **minutes**.

4. Reading "O'clock" Times

- "O'clock" means exactly on the hour.
- **Rule:** When the **Long Hand (Minute Hand)** points straight up to the **12**, it is "o'clock".
- To know which hour it is, we look at the **Short Hand (Hour Hand)**.
 - If the short hand points to 3, and the long hand is on 12, the time is **3 o'clock**.
 - If the short hand points to 9, and the long hand is on 12, the time is **9 o'clock**.

12
11 1
10 2
9 3
8 4
7 5
6

Short hand on 3 = 3 O'Clock

EVALUATION

1. How many days are there in one week?
2. What day comes after Monday?
3. When the long hand is on 12 and the short hand is on 5, what time is it?
4. What part of the day do we usually eat breakfast?
5. Draw the face of a clock showing **2 O'clock**.

ASSIGNMENT

1. Write the days of the week in order, starting with Sunday.
2. Draw a clock face. Show where the hands should be to tell **6 O'clock**.
3. What day is today? Write it down.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions based on the topics covered from Week 1 to Week 4.

CONTENT

Assessment of Weeks 1–4

- **Fractions:** Understanding and identifying **half** ($\frac{1}{2}$) and **quarter** ($\frac{1}{4}$).
- **Measurement:**
 - Length (Long/Short, Longer/Shorter, measuring with hand spans).
 - Weight (Heavy/Light, Heavier/Lighter, balance scale).
- **Time:** Parts of the day, days of the week, reading **O'clock** times on a clock.

EVALUATION

- Administer the written and practical Mid-Term Test.

ASSIGNMENT

- Nil.

WEEK 6: MID-TERM BREAK

OBJECTIVES

By the end of the week, pupils should be able to:

1. Rest and refresh for the second half of the term.

CONTENT

Holiday/Rest

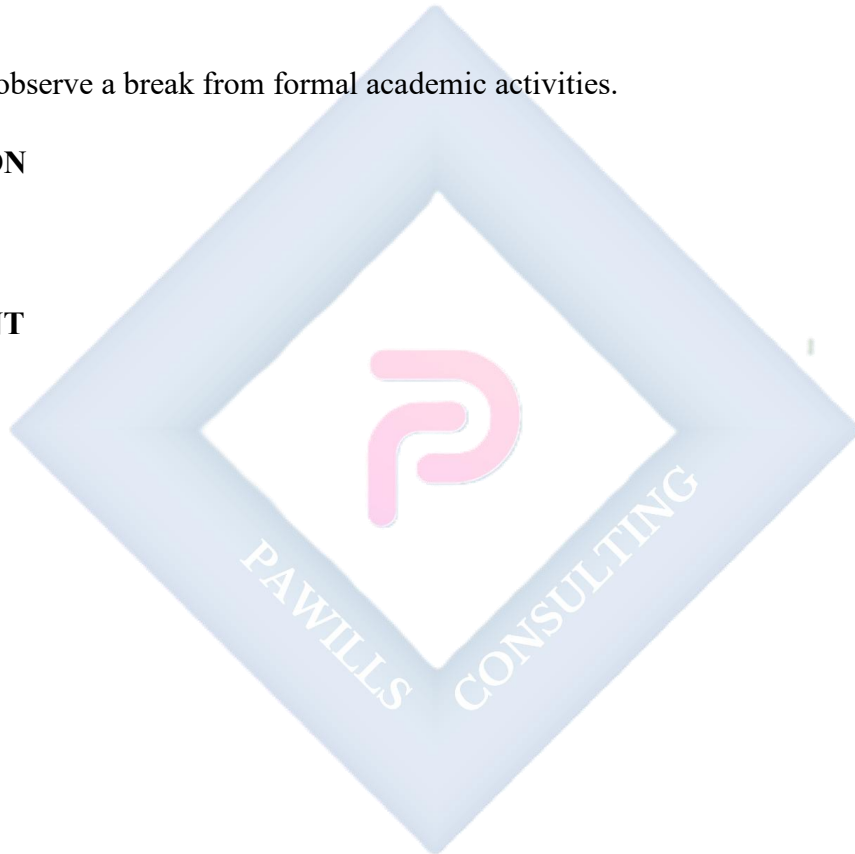
- Pupils observe a break from formal academic activities.

EVALUATION

- Nil.

ASSIGNMENT

- Nil.



WEEK 7: THREE DIMENSIONAL SHAPES

OBJECTIVES

By the end of the lesson, pupils should be able to:

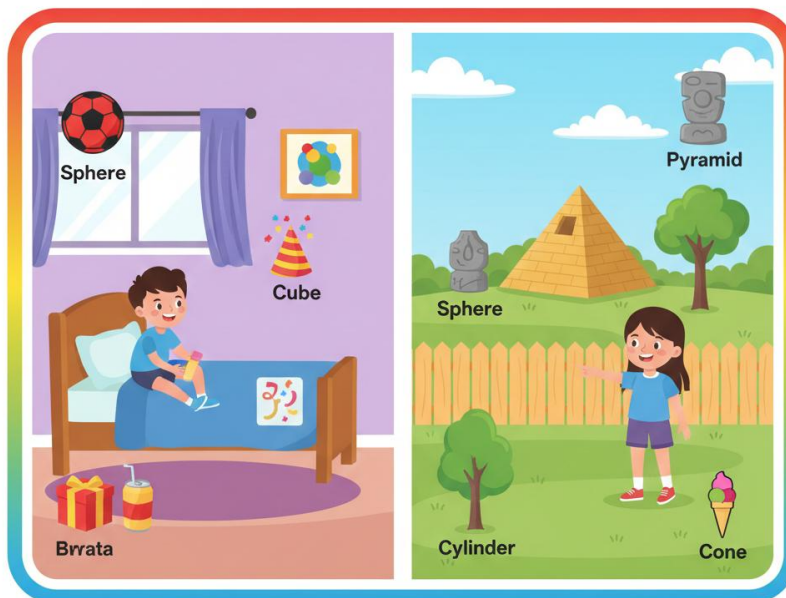
1. Identify and name 3D shapes: Cube, Cuboid, Sphere, Cylinder, Cone.
2. Match 3D shapes to real-life objects (e.g., Sphere to a ball).
3. Mention simple properties of 3D shapes (faces, edges).

CONTENT

1. Identification of 3D Shapes

- **3D shapes** are also called **solid shapes**. They are not flat; they are "fat" and you can hold them in your hand. They take up space.
- **Cube:** A box with 6 square faces that are all the same size. Example: a dice, a sugar cube.

3D Shapes All Around Us!



Sphere



Cone



Cone



Cylinder



Cone

- **Cuboid:** A box with 6 rectangular faces (like a stretched cube). Example: a book, a brick, a matchbox.
- **Sphere:** A perfectly round ball shape. Example: a football, a globe, an orange.
- **Cylinder:** A shape with two flat, circular ends and a curved surface. Example: a tin of milk, a drum, a pipe.
- **Cone:** A shape with a circular base and a pointed top (apex). Example: an ice-cream cone, a party hat, a traffic cone.

2. Properties of 3D Shapes (Faces, Edges, Corners)

- **Face:** The flat or curved surface of a 3D shape. A cube has **6 faces**.
- **Edge:** The line where two faces meet. A cube has **12 edges**.
- **Corner/Vertex:** The point where edges meet. A cube has **8 corners**.

3. Finding Shapes in the Environment

- We can play "Shape Hunt" in the classroom.
- **Cuboid:** The door, the window frame, a cupboard.
- **Cylinder:** A pencil (if not sharpened), a bottle, a pillar.
- **Sphere:** A bead, some toys.
- **Cube:** Some building blocks, a tissue box (if square).

EVALUATION

1. Name the shape that looks like a ball.
2. Name the shape that looks like a tin of milk.
3. How many flat faces does a cube have? (Let's count together).
4. Is a party hat shaped like a cylinder or a cone?
5. Name one object in your home that is a cuboid shape.

ASSIGNMENT

1. Draw a picture of a **Cylinder** and a **Cone**.
2. Write the name of the 3D shape of a Maggi cube.

3. Find one **Sphere** in your house (e.g., a ball, some fruit) and write its name.



WEEK 8: TWO DIMENSIONAL SHAPES

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify and name 2D shapes: Circle, Triangle, Square, Rectangle, Oval.
2. Describe properties of 2D shapes (sides and corners).
3. Draw basic 2D shapes correctly.

CONTENT



1. Identification of 2D Shapes

- **2D shapes** are **flat shapes**. They have length and width but no thickness, like a drawing on paper.
- **Circle**: A round shape with one curved side. It has no corners. Example: a coin, a clock face.

- **Triangle:** A shape with **3 straight sides** and **3 corners**. Example: a slice of pizza, a road sign.
- **Square:** A shape with **4 equal straight sides** and **4 corners** (all are right angles). Example: a chessboard square, a window pane.
- **Rectangle:** A shape with **4 straight sides**. Opposite sides are equal. It has **4 corners**. Example: a door, a sheet of paper, a blackboard.
- **Oval:** A stretched-out circle, like the shape of an egg or a human face.

2. Properties of 2D Shapes (Sides and Corners)

- **Side:** A straight or curved line that forms part of the boundary of a shape.
- **Corner/Vertex:** The point where two sides meet.
- **Summary:**
 - **Triangle:** 3 sides, 3 corners.
 - **Square:** 4 equal sides, 4 corners.
 - **Rectangle:** 4 sides (2 long, 2 short), 4 corners.
 - **Circle:** 1 curved side, 0 corners.
 - **Oval:** 1 curved side, 0 corners.

3. Sorting and Drawing Shapes

- We can sort shapes by the number of sides: shapes with 3 sides, shapes with 4 sides, shapes with no sides (curved).
- Use a ruler to help draw shapes with straight sides (triangle, square, rectangle).

EVALUATION

1. How many sides does a triangle have?
2. Which shape looks like an egg?
3. Draw a square in your book.
4. Does a circle have any corners?
5. What is the main difference between a square and a rectangle? (Square has all sides equal; rectangle has opposite sides equal).

ASSIGNMENT

1. Draw a **Rectangle** and color it blue.
2. Draw a **Triangle** and color it yellow.
3. Count the number of corners on a square and write the number.



WEEK 9: TWO DIMENSIONAL SHAPES (CONTINUED) & PATTERNS

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Create simple patterns using 2D shapes.
2. Identify the 2D shape found on the face of a 3D shape.
3. Explain symmetry as being the "same on both sides".
4. Combine shapes to form a picture.

CONTENT



1. Creating Patterns with Shapes

- A **pattern** is a sequence that repeats in a predictable way. Patterns are everywhere (clothes, tiles, nature).

- We can make patterns using shapes, colors, or objects.
- **Example 1 (Shape Pattern):** Circle, Square, Circle, Square, Circle, ? (What comes next? Square).
- **Example 2 (Color & Shape):** Red Triangle, Blue Triangle, Red Triangle, Blue Triangle, ? (What comes next? Red Triangle).

2. Relationship Between 2D and 3D Shapes

- The flat surfaces (faces) of 3D shapes are 2D shapes!
- If you stamp the bottom of a **Cube** in paint, the print it leaves is a **Square**.
- If you stamp the end of a **Cylinder**, the print is a **Circle**.
- The face of a **Cuboid** is usually a **Rectangle**.

3. Introduction to Symmetry

- **Symmetry** means that one half of a shape or picture is a mirror image of the other half. If you fold it on the line of symmetry, both halves match perfectly.
- A butterfly's wings are symmetrical.
- A circle has many lines of symmetry. A square has 4.
- **Activity:** Fold a piece of paper, cut a simple shape out of the folded edge, and open it up to see a symmetrical shape.

4. Combining Shapes to Make Pictures

- We can use simple 2D shapes as building blocks to draw objects.
- **A House:** A square for the house + a triangle for the roof.
- **A Car:** Circles for wheels + rectangles for the body.
- **A Tree:** A rectangle for the trunk + a circle for the leaves.

EVALUATION

1. Complete this pattern: Square, Star, Square, Star, Square, ?.
2. What 2D shape is on each face of a cube?
3. If you put a triangle on top of a square, what common picture does it look like?

4. Draw a line to show where you could fold a circle so that both halves match (a line of symmetry).
5. What comes next in this letter pattern: A, B, A, B, ?.

ASSIGNMENT

1. Draw a simple repeating pattern using Triangles and Circles.
2. Draw a house using a Triangle and a Square. Color it.
3. Trace the bottom of a cup or bottle on a paper. What 2D shape did you get?



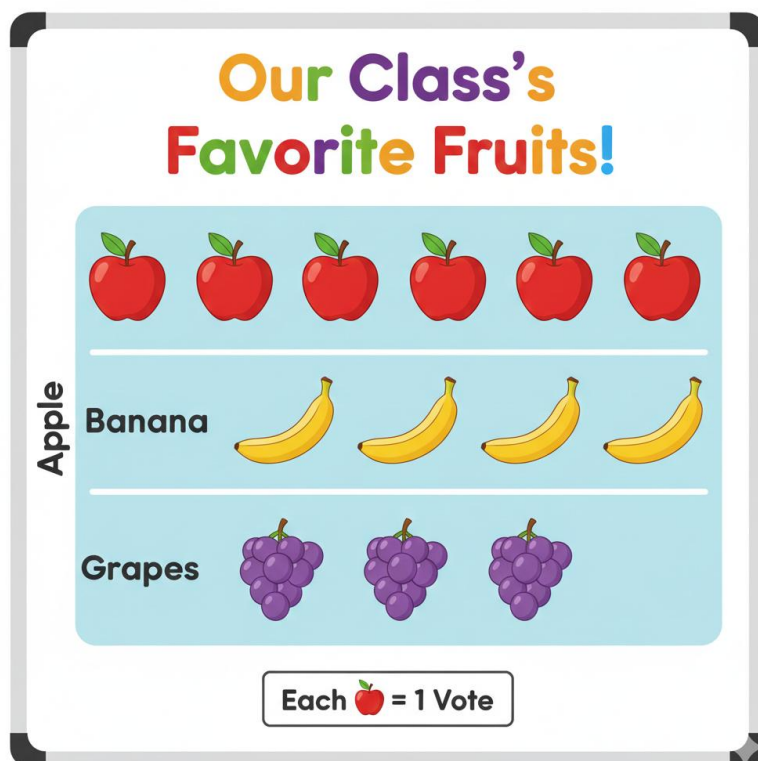
WEEK 10: DATA COLLECTION (INTRODUCTION)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define data collection as gathering information.
2. Use tally marks to count objects.
3. Read and interpret a simple pictograph.
4. Organize data into a simple table.

CONTENT



1. Understanding Data Collection

- **Data** is a collection of facts or information. We collect data to answer questions.
- **Data Collection** means gathering this information in an organized way.
- **Example Question:** "What is the favorite color of pupils in our class?" To answer, we must collect data by asking everyone.

2. Using Tally Marks

- **Tally marks** are a quick way to count and record data as you go. We group them in fives for easy counting.
- **How to make tally marks:**
 - 1 = |
 - 2 = ||
 - 3 = |||
 - 4 = ||||
 - 5 = ||||| (Four vertical lines crossed by a diagonal fifth line). This is a "bundle of five."
- **Example:** Counting 7 pupils: |||| | (This is $5 + 2 = 7$).

3. Collecting Class Data (A Survey)

- Let's collect real data. Ask the class: "Do you have a pet? Yes or No?"
- As pupils raise hands, make tally marks for "Yes" and "No" on the board.
- Count the tally marks to see the total for each answer.

4. Pictographs

- A **pictograph** uses pictures or symbols to show data. Each picture stands for one item or one person.
- **Example:** Favorite Fruit
 - Apple: 😊 😊 😊 (3 pupils like apples)
 - Banana: 😊 😊 (2 pupils like bananas)
- It is easy to see which has more pictures, so which is more popular.

EVALUATION

1. What is **data**?
2. Draw tally marks for the number 3.
3. Look at this simple pictograph: Blue: ♥♥♥♥. How many children chose Blue?

4. Draw tally marks for the number **5**.
5. In our fruit pictograph, which fruit was more popular: Apple (3) or Banana (2)?

ASSIGNMENT

1. Count the number of **windows** in the main room of your house. Record the number using tally marks.
2. Ask **3 friends** what their favorite color is. Write down their answers.
3. Draw a simple pictograph showing 2 balls and 4 cups. (Use a simple symbol for each).



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from Fractions, Measurement, Shapes, and Data.
2. Demonstrate readiness for the end-of-term examination.

CONTENT

Review of All Topics

- **Fractions:**
 - $\frac{1}{2}$ (**Half**): One of two equal parts.
 - $\frac{1}{4}$ (**Quarter**): One of four equal parts.
- **Measurement:**
 - **Length:** Long/Short, Longer/Shorter, measuring with hand spans.
 - **Weight:** Heavy/Light, Heavier/Lighter, the balance scale rule (heavy side goes down).
 - **Time:** Parts of the day, Days of the week, Reading **O'clock** times (long hand on 12).
- **Shapes:**
 - **3D (Solid):** Cube, Cuboid, Sphere, Cylinder, Cone. Real-life examples.
 - **2D (Flat):** Circle, Triangle, Square, Rectangle, Oval. Sides and corners.
 - **Patterns & Symmetry:** Creating simple repeating patterns.
- **Data Collection:** Using **tally marks** and understanding simple **pictographs**.

EVALUATION

- Oral quiz: "Name a 3D shape shaped like a ball." "What is half of something?"
- Practical: Show a clock set to an 'o'clock' time for pupils to read.
- Drawing: Draw a triangle and a rectangle on the board.

ASSIGNMENT

- Study all notes from Week 1 to Week 10 in preparation for the end-of-term examination.

